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INTRODUCTION

The seed of today's mobile computing was implanted long back in 1897 since G. Marconi invented the radio's ability to provide continuous contact with ships in the English Channel. The seed was grown up to a sapling and finally turned into a huge tree. This is due to the improvements in digital and RF circuit fabrication, a large-scale circuit integration and other miniaturization technologies. The mobile computing spreads its branches providing high speed text and multimedia data services as well as extends its capacity in terms of subscriber accommodation. It enables today's cheap, improved, portable and enriched services available for the entire population. This reality is of many years' effort in different wings of technology.

1.1 HISTORY

Prior to the analog cellular system was in operation (Chicago, 1983), the mobile radio systems were based on trunking principle. Designers provided optimal number of channels for sharing among the users. The available frequency spectrum (150–450 MHz) was divided into a number of frequency channels. The channels were shared by the users. Such a system used single high-powered transmitter installed on a large tower. The transmitter was powerful enough to cover a distance of 50 km. However, the concerns of those systems were:

- the mobile receivers (generally installed in automobile) were of large size and therefore, difficult to carry.
- a very limited number of channels were used. Even for call termination, a mobile station had to compete for a channel.

- the call blocking probability was high.
- the coverage of a network was limited.
- the number of simultaneous calls in progress was very restricted.

In spite of radio's (wireless) tremendous ability to communicate anytime anywhere, the concept of wireless communication was not so popular in a mass scale. During the 1960s Bell Laboratories developed the cellular concept. A major breakthrough took place with the introduction of frequency reuse mechanism in cellular operation. Simultaneously, the advancement in IC technology, digital signal processing and battery technology aided the sharp growth in mobile and personal communication services. Such systems were primarily analog and were best known as 1st generation mobile systems.

Around late 1980s the Wireless Local Area Network (WLAN) technology, an alternative to wired LAN, was launched. The two prominent standards for WLAN emerged were the IEEE 802.11 in the US and the High Performance European Radio LAN (HIPERLAN) in Europe. In 1990, the first digital cellular communication was implemented in Europe. It operates in 935–960 MHz and in 890–915 MHz bands. The systems using digital cellular technology are perceived as the 2nd generation mobile or Personal Communication Services (PCS) systems. The notable among the 2nd generation systems are the Digital European Cordless Telephone (DECT) and Digital Communication Systems (DCS, 1800 MHz version of GSM) introduced in Europe, IS-95 in North America, and Personal Digital Cellular (PDC) in Japan.

The initial focus of the 2nd generation mobile systems was on circuit-switched voice and low bit rate data services. In circuit-switched connection, a dedicated connection is established for the entire duration of a call. In cellular network, circuit-switched voice demands a dedicated radio channel between a handset (mobile station) and its nearest network tower (base station) and a dedicated phone line between the terminal point of the network (mobile switching centre) and telephone network (public switch telephone network), even when the mobile station moves to a different place under the new network tower.

Over the years, the demand to have higher data rates increases for applications such as Internet access, various standards like General Packet Radio Service (GPRS) for GSM, Cellular Digital Packet Data (CDPD), High Speed Circuit-Switched Data (HSCSD), etc. These are considered as gateway to next generation wireless access and, therefore, referred to as 2.5 generation standards/systems. However, the 2nd and 2.5 generation systems are lacking in

- Interoperability among cellular, cordless, satellite, LANs, etc.
- Handling multimedia services along with voice, data and Internet.
- Global mobility—that is, global seamless roaming.
- Quality of Service (QoS).

Although the first rush to wireless was for voice transmission, attention was also paid to data services demanding the wireless Internet. Two technologies/standards, Mobile IP and Wireless Application Protocol (WAP), were proposed to provide application level support for wireless networking.

The Mobile IP was conceived around 1992 and standardized in 1996. It was targeted to enable seamless roaming for Internet connected devices. Simultaneously, around 1997 the WAP was introduced and then implemented in 2000. With the WAP a wireless service provider could provide Internet services to mobile devices over the same voice network. The evolution of mobile communication services is recorded in Figure 1.1.

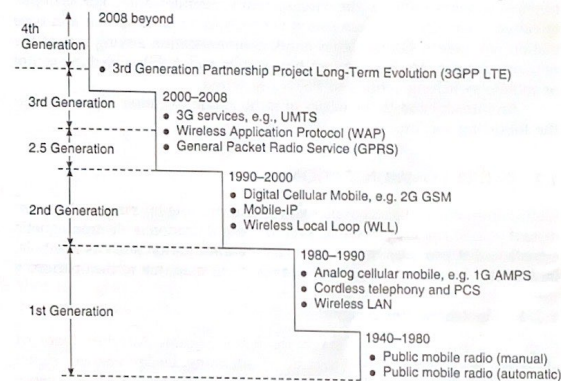


Figure 1.1 Evolution of mobile communication services.

To meet the challenges of 2nd generation mobile systems, the International Telecommunication Union (ITU) has developed a standard IMT-2000 and initiated the attempt towards development of the 3rd generation mobile systems. The 3rd generation systems accommodate the existing wireless access systems (that are the harvest of many efforts from different sectors) under one umbrella. The Universal Mobile Telecommunication Systems (UMTS), proposed by European Telecommunications Standard Institute (ETSI), is developed at par with the IMT-2000. It supports global roaming facility as well as the high-speed data services for different QoS traffic classes ranging from the ordinary background data (e.g. e-mail) to the most essential conversational data (e.g. voice) or even the highly prioritized multimedia services (e.g. video on demand).

The higher data rate is projected from 4th generation mobile systems. It will be able to provide an all IP network where voice, data and streamed

multimedia can be given to users on an anytime anywhere basis, and at higher data rates than 3rd generation. The international telecommunications regulatory and standardization bodies are working for commercial availability of 4G networks around 2012–2015. One of the technologies which are being considered as pre-4G is 3rd Generation Partnership Project Long Term Evolution (3GPP LTE). The LTE project is not a standard, but it will result in the extensions and modifications of the UMTS system. It is to improve the UMTS mobile phone standard to cope up with future technology evolutions.

In all mobile communication networks, at least one hop (if not all hops) of communication is the wireless radio communication. For example, in cellular network, communication between the mobile station and base station is wireless. On the other hand, communication among rest of the network components, e.g. between base station and mobile switching centre or mobile switching centres to PSTN are wired.

An introduction to the basics of radio communication is provided in the following section.

1.2 RADIO COMMUNICATION

Radio communication is a popular way of transmitting information between distant places. The unit of wireless radio communication is electromagnetic signals and it provides means to exchange information such as symbols, intelligence between two points—that is, from a source to destination.

1.2.1 Electromagnetic Signal

Light is a form of electromagnetic wave/signal. Another form of electromagnetic signal is used in radio broadcasting. Electromagnetic signal is oscillatory in nature. The graph displayed in Figure 1.2 is a snapshot representing the strength of a signal. The distance between two wave crests is the wavelength of this signal. The signal traverses at the velocity of light.

A signal is qualified by its frequency of oscillation—that is, the number of oscillations per second. The usual frequency we get from the electric currents in our household connection is about 50/60 cycles per second. In radio (AM) broadcasting, frequency of the electromagnetic signal used is in the order of 10^6 cycles per second. The frequency used for FM radio/TV is more than the radio broadcast and short waves/radar frequency is more than the frequency used for FM radio/TV. For reference, it can be stated that the further increase in the frequency of oscillation make electromagnetic signal visible by human eyes. The visible frequency is in the range of 5×10^{14} to 5×10^{15} cycles per second.

The unit of frequency measure is Hertz (Hz); 1 Hz represents the frequency of one cycle per second. The megahertz (MHz) is the 10^6 Hz and gigahertz (GHz) implies 10^9 Hz. That is, 1 MHz indicates that the

signal completes one million (10^6) cycles per second and 1 GHz implies 10^9 cycles per second.

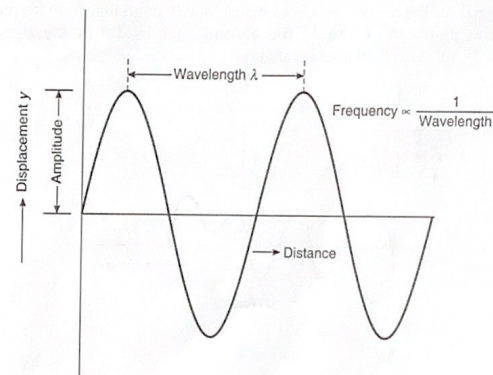


Figure 1.2 A typical electromagnetic signal.

The variable quality of the electromagnetic waves could also be the wavelength. The frequency is inversely proportional to the wavelength. That is, higher frequencies have the shorter wavelengths, while lower frequencies indicate larger wavelength. The frequency (f) can be represented as $f = v/\lambda$, where v = phase velocity of wave and λ = wavelength. In vacuum, v is the velocity of light. The radio waves can be of wide range of wavelengths. The regular radio broadcasting wavelength is about 500 metres. The wavelength of short-wave/radio wave is in the order of millimetre. Further, the frequency $f = 1/T$, where T (period) is the time between two consecutive occurrences of the event in wave propagation. For example, if period T of a signal is 1s (10^0 s), then the frequency $f = 1$ Hz (10^0), and if $T = 1$ ms (10^{-3} s), $f = 1$ kHz (10^3 Hz).

Analog vs digital signal

The signal can be either analog or digital. An analog signal is one in which the signal intensity varies in a smooth fashion (continuous) over time. On the contrary, a digital signal is one in which the signal intensity maintains a constant level (discrete) for some period of time and then changes to another constant level. Figure 1.3 shows a sample of analog and digital signals.

¹ The phase velocity of a wave is the rate at which the phase of the wave propagates in space.

Spectrum/bandwidth

The electromagnetic waves cover a wide frequency range. The spectrum of a signal is the range of frequencies that it contains. If a frequency spectrum ranges from f to $3f$, the absolute bandwidth of the signal is $3f - f = 2f$, the width of the spectrum.

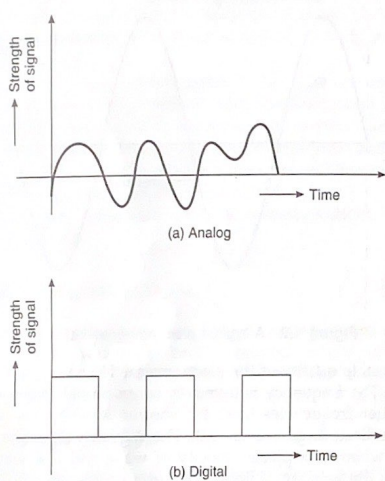


Figure 1.3 Signal.

Many signals have an infinite bandwidth. However, most of the energy contains within a relatively narrow band of frequencies. This narrow band is referred to as the *effective bandwidth* or simply *bandwidth*. In radio, the width of frequency spectrum can impart more information in communication process.

Characteristics of electromagnetic signals

Table 1.1 shows the spectrum of electromagnetic signals with common nomenclature. The signals have different characteristics and abilities for communication. The key considerations are: how far the waves can go, through what these can go through and how much data these can carry. All these have an impact on how useful a signal will be for different types of wireless communication. In the current context, the main concern is on radio portion (a few cycles to 10^9 cycles per second) of the spectrum (Refer to Figure 1.4) that is used in mobile computing.

Table 1.1 Radio Frequency/Bandwidth

Frequency	Band	Type of radiation
< 30 kHz	Very Low Frequency (VLF)	Long radio wave
30–300 kHz	Low Frequency (LF)	Long radio wave
300 kHz–3 MHz	Medium Frequency (MF)	Long radio wave
3–30 MHz	High Frequency (HF)	Short radio wave
30–300 MHz	Very High Frequency (VHF)	Short radio wave
300 MHz–3 GHz	Ultra High Frequency (UHF)	Short radio wave
3–30 GHz	Super High Frequency (SHF)	Microwave
> 30 GHz	Extremely High Frequency (EHF)	Infrared ... Gamma rays

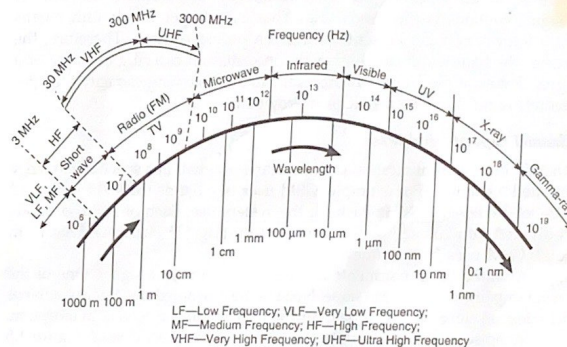


Figure 1.4 The electromagnetic spectrum.

Range of transmission

High frequencies (HF) are reflected by the ionosphere and bounce back and forth between the ionosphere and the earth's surface. As a result the HF signal can convey information/data to thousands of kilometres. Further, a HF signal can penetrate obstructions such as buildings, mountains, etc. Such a frequency is typically used by the radio stations to broadcast programs worldwide.

The higher frequencies such as SHF and EHF normally operate on a line-of-sight (LOS) basis. In LOS, the receiver receives the signal from the transmitter on a straight line. These frequencies do not bend around the earth's surface like HF and, therefore, require repeaters, every few tens of kilometres, to establish links between the nodes.

Microwaves (SHFs) can pass right through some objects but are blocked by relatively solid objects like buildings and metal structures. The light waves behave like microwaves but light waves are even more

restricted in distance travelled and comparatively less powerful to penetrate objects.

The range of transmission depends not only on the characteristics of a signal but also on the parameters such as power of the transmitter, sensitivity of receiver, antenna types, antenna height, quality of the transmission medium used, method used to send information, amount of interference present, atmospheric conditions, etc.

Information-carrying capacity

There is a direct relationship between the information-carrying capacity of a signal and its bandwidth. The information-related modifications made to the signal are normally tied to its frequency. The range of frequencies a signal contains is the bandwidth. That is, a larger bandwidth means more information can be sent for a given period of time. Therefore, the greater the bandwidth, the higher the information-carrying capacity of a signal. For example, in broadband, information-carrying capacity (bits per second) is far more than that in narrowband.

Channel capacity and noise

An individual communication path that carries signals at a specific frequency is called a channel. For example, GSM uses two bands (890–915 MHz and 935–960 MHz) of 25 MHz each for the system use. Each of 25 MHz bands is divided into 200 kHz wide channels resulting 125 channels. That is, in total, GSM uses 250 channels.

A variety of impairments can distort or corrupt a signal. One of the major impairments is due to noise. Noise is the unwanted signal that combines to a desired signal and distorts it during transmission as well as at reception.

A noise affects and limits the data rate in transmission. Figure 1.5 shows how a transmitted signal is affected by noise in transmission medium. The 'channel capacity' qualifies such effect of noise. The maximum rate at which the data can be transmitted over a channel, under given conditions, is referred to as the 'channel capacity'. It is the upper bound on the amount of information that can be reliably transmitted over a communication channel. On the other hand, 'signal strength' is the magnitude of electric field at a reference location around the transmitting antenna.

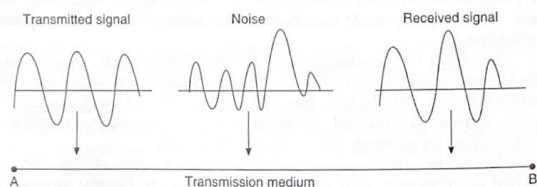


Figure 1.5 Signal distortion due to noise.

The greater signal strength improves the ability to receive data correctly in the presence of noise. The signal-to-noise ratio (SNR), the ratio of power in a signal to the noise power (corrupting the signal at a particular point of transmission), is the measure of data received correctly and is defined as

$$SNR = \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right) = \left(\frac{A_{\text{signal}}}{A_{\text{noise}}} \right)^2$$

P refers average power and A is the root mean square amplitude.

It is usually expressed in terms of logarithmic scale. In decibel it is 10 times the logarithm of power ratio. That is,

$$SNR_{dB} = 10 \log_{10} \left(\frac{\text{Signal power}}{\text{Noise power}} \right)$$

For example, in a cellular system, if the signal power/noise power = 96.2, the SNR = 20 dB. A high value of SNR means signal received is of high quality. The SNR is also referred to as the signal-to-interference ratio (S/I).

1.2.2 Full-duplex Radio Communication

Based on the direction of communication, a radio communication system can be referred to as either simplex or duplex. A simplex system allows

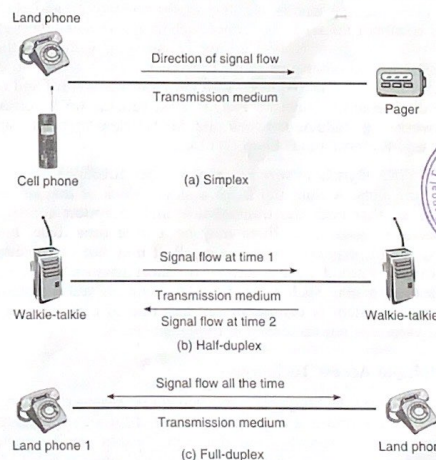


Figure 1.6 Radio communication.

communication in one direction. An example of simplex system can be the pager. A pager receives message but cannot reply/acknowledge. A duplex system, on the other hand, allows two-way communication between two points. In half-duplex radio system, the two-way communication cannot occur simultaneously. The same half-duplex channel can be used for transmission as well as reception; but at any point of time either transmission or reception can take place. Walkie-talkie is an example of half-duplex system. Whereas, the full duplexing is a mechanism by means of which simultaneous transmission and reception is possible in a communication system. In other words, communication is possible in both directions simultaneously. The conventional telephone system is the right example of full-duplex system, where simultaneous exchange of dialogue is possible. Figure 1.6 shows various types of radio communication.

In wireless systems, full duplexing is implemented either by providing two separate channels (one for transmission and the other for reception) or by providing two adjacent time slots on a single radio channel. The former method for achieving full-duplex communication is called the Frequency Division Duplexing (FDD) and the latter one is called the Time Division Duplexing (TDD).

FDD: In case of FDD, signals are differentiated based on the frequencies. Two distinct set of frequencies are assigned for this purpose. Each such set of frequencies corresponds to a simplex channel in FDD and, therefore, a duplex channel consists of two simplex channels. In a cellular network, a device called duplexer is used in mobile station as well as in the base station allowing simultaneous radio transmission and reception on the duplex channel pair. The frequency split between the forward and reverse (backward) channels is constant. FDD is most suitable for wireless wide area network, e.g. cellular network and for wireless metropolitan area network, e.g. Wireless Local Loop (WLL).

TDD: In TDD, signals in forward and reverse directions are assigned separate time slots. A time slot is for a single block of frequencies and kept small so that both the transmission and reception appear to be simultaneous to a user. As there may be a little time delay between transmission and reception, however small, it may not be considered a true full-duplex method. TDD is suitable for short distance and low power communication system such as cordless telecommunication system. The cost of such a system is comparatively less due to a fewer wires and simplified receivers required for the implementation.

1.2.3 Multiple Access Techniques

The 'multiple access' technique is a mechanism by means of which many users can share the frequency spectrum allocated for an application. The systems specify how signals from different sources can be combined efficiently for transmission over a given radio frequency band. At the receiving end, the signals are then separated out ensuring minimum mutual

interference. There are mainly three multiple access techniques—Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA), and Code Division Multiple Access (CDMA). FDMA system separates the total available bandwidth into several non-overlapping smaller bands/channels. Each channel has the ability to support a user. TDMA system splits users into an available pair of channels, but it also assigns each user an available time-slot within that channel. In each slot, only one user is allowed to either transmit or receive. With CDMA, unique digital codes, rather than the separate radio frequencies/channels, are used to differentiate users. A specific code is assigned to each user and only that code can demodulate the transmitted signal. Each user utilises the entire block of allocated spectrum space to carry the message. An elaborate discussion on these techniques are provided in Chapter 3 as the part of digital communication.

1.2.4 Operational Modules for Radio Communication

In this subsection, the major tasks involved in radio communication are identified. Figure 1.7 shows the operational modules in both the transmitter and receiver ends. When a signal, i.e. voice, video, etc. is generated for communication, it is processed in several phases prior to transmission. At first the analog signal is transformed into digital by the source coder. The channel coder detects and/or corrects the errors due to transformation. However, all the errors cannot be corrected by the channel coder. The effect of such errors is minimized by the interleaver. Finally, the signal is given upshift of frequency so that it can be transmitted over the radio path in an efficient manner. When the signal is received at the receiver, a reverse process is applied to extract the original signal.

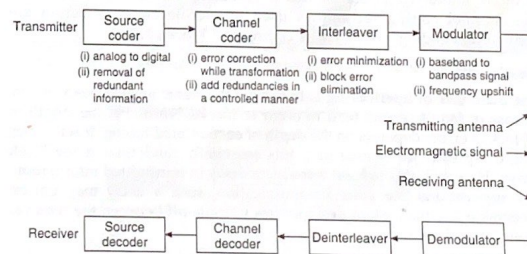


Figure 1.7 Operational modules for radio system.

Source coding

In this phase the input signal to be transmitted (voice, video, etc.) is transformed from analog to digital. In wireless communication, there exists

a scarcity of bandwidth. However, for applications like voice communication, high quality service is needed. Therefore, the challenge in such speech coding system is to ensure high quality speech transmission exploiting the least possible channel capacity. It demands a trade-off between these two conflicting requirements. In order to maximize spectral efficiency², the more efficient voice coding methods are introduced in cellular systems. Once analog signal is converted to digital, the encoded data bits, called source bits, carry information. Due to a typical coding mechanism, some of the source bits have more impact than others and therefore need to be protected from errors. On the other hand, to save the bandwidth, bits carrying redundant information should be removed prior to transmission. So, the removal of redundant information is also the task of a source coder.

Channel coding

After removal of redundancies by the source coder, in a radio system, the source bits pass through the channel coding process. It improves quality of transmission by protecting source data from errors caused by the events such as multipath fading (transmitted signals take multiple paths to reach to the receiver causing interference between multiple versions of the transmitted signal having differences in amplitude and phase received at slightly different times), Doppler shift (due to movements), etc. The data/information is protected from errors through introduction of redundancies in a controlled manner. However, the errors such as block error caused by fading cannot be efficiently corrected by the channel coding schemes and demand interleaving. A block error results in alteration of a sequence of bits (block of bits) from the transmitted bits. If the alteration can be spread within the whole bit stream, chances of loss of message are reduced. Interleaving does the task of spreading the loss of bits. Some of the channel coding techniques used in mobile communication are convolution coding, parity check coding and block coding.

Interleaving

The main task of interleaving is to protect the signal against block errors. In case of fading, errors tend to occur in blocks. Normally, the length of a block of errors depends on the depth of encountered fading. It scrambles and/or spreads the source data bits essentially randomizing the block errors. However, this process introduces delay in transmitted information. For applications like voice communication, such a delay may not be acceptable. So, the system designer has to trade-off between the removal of errors and delay.

Modulation

It is necessary that the source data needs to be transformed in a manner suitable for transmission in wireless communication system. Wireless

² Spectral efficiency is the amount of information that can be transmitted over a given bandwidth.

medium allows only analog transmission. If digital source data has to be transmitted over such a medium, the modulation is required. The modulation translates a source or baseband message signal to a bandpass signal (whose frequency content is concentrated in a narrow-band) at frequencies very high compared to a baseband frequency. The source or baseband message signal is called modulating signal and the generated bandpass signal is the modulated signal. The high frequency signal that used to translate the baseband message signal to the bandpass signal is called the carrier signal (Figure 1.8(a)).

There are various reasons for which the baseband signal cannot be directly transmitted in a wireless medium. The important one is the antenna height. It must be in order of magnitude of the signal's wavelength that is to be transmitted. For example, even for 1 MHz signal, antenna height must be in order of hundred metres or so. Therefore, the modulation phase basically performs two tasks—(i) digital modulation, i.e. translation of source encoded and channel encoded digital data into square wave binary pulses. This is the baseband signal if presented in frequency domain, (ii) frequency shifting. Because of signal attenuation, impractical antenna height, etc., it is necessary to shift the baseband signal spectrum to reside at a much higher centre frequency of the baseband signal. The three basic methods for the digital modulation are amplitude shift keying (ASK), frequency shift keying (FSK) and phase shift keying (PSK). Similarly, there are three analog modulation techniques—amplitude modulation (AM), frequency modulation (FM) and phase modulation (PM). The digital version of amplitude modulation (AM), frequency modulation (FM) and phase modulation (PM) are the amplitude shift keying (ASK), frequency shift keying (FSK) and phase shift keying (PSK) respectively.

AM: Amplitude modulation alters the amplitude of a carrier signal so that its amplitude matches with the original, i.e. baseband modulating signal (Figure 1.8(a)). That is, carrier is modulated such that its amplitude varies with the amplitude of the baseband modulating signal.

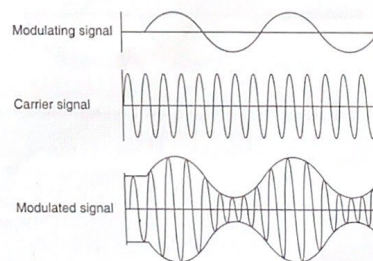


Figure 1.8(a) Amplitude modulation.

The frequency and phase of the carrier signal remain unaltered. It must be ensured that the frequency of the carrier must be much higher than the highest frequency in the baseband signal. The demodulation process at the receiving end filters out the carrier. The bandwidth required in modulated signal is twice of the bandwidth of a modulating signal.

AM works by varying the strength of the transmitted signal relating to the information being sent and suffers from power variation in the output.

ASK: The input to such a modulation process is the binary bit streams. The two binary values 1 and 0 are represented by two different amplitudes. Figure 1.8(b) shows the transformation of input bit stream 1001. In this example, the binary 0 is represented by amplitude 0. Here the carrier signal strength is varied in modulated signal. The frequency and phase of the carrier remains constant but amplitude is changed to match with the input binary signal (repetition). However, various interferences in radio propagation such as multi-path propagation, noise, etc. affect amplitude greatly. So, ASK is normally not preferred in wireless radio transmission.

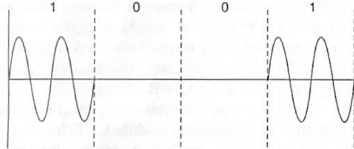


Figure 1.8(b) Amplitude shift keying.

FM: The frequency of the carrier signal is altered to carry the content of the modulating signal (Figure 1.9(a)). The other two parameters, i.e. amplitude and phase of the carrier are kept constant. As the carrier signal's

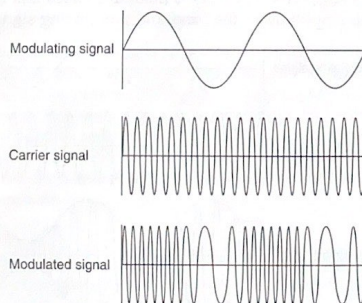


Figure 1.9(a) Frequency modulation.

amplitude remains unchanged, the FM is more immune to noise than AM and improves the overall signal-to-noise ratio (SNR) in the system. Unlike AM, the power output in FM is constant.

Frequency modulation is achieved by simply mixing the modulating and carrier signals. It follows the amplitude of the modulating signal. The range of frequency deviation in the modulated signal is proportional to the amplitude of modulating signal. For example, in modulation, if the carrier frequency is 1000 MHz and the modulating signal's amplitude is 20, then for 40 kHz frequency shift per unit amplitude, the modulated signal's frequency variation will be in between $(1000 \text{ MHz} - 20 \times 40 \text{ kHz})$ and $(1000 \text{ MHz} + 20 \times 40 \text{ kHz})$, i.e. 999.2 MHz and 1000.8 MHz.

The number of instances of such variation in frequency within the modulated signals are equal to the frequency of modulating signals. It means if the modulating signal frequency for the earlier example is 2 kHz, then the frequency variation (from 999.2 to 1000.8 MHz) in modulated signal will occur at the rate of 2000 times per second.

Further, the bandwidth for FM is 10 times of the bandwidth of modulating signal.

FSK: The simplest form of FSK is the binary FSK (BFSK). Instead of representing the input binary values by two different amplitudes, the BFSK represents 1 and 0 by the two different frequencies that are close to the carrier frequency. The FSK representation of example bit stream 1001 is shown in Figure 1.9(b). It can be noted that the peak amplitude and the phase remain constant. The FSK is more immune to noise than the ASK.

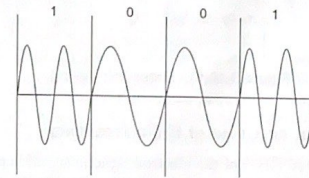


Figure 1.9(b) Frequency shift keying.

PM: In phase modulation, the carrier signal's amplitude and frequency remain constant after modulation. It is either advanced or retarded in its phase cycle (phase shift) by the modulating signal (Figure 1.10(a)) and the amount of phase shift is proportional to the amplitude of the modulating signal. The PM is less prone to the noise interference.

PSK: In phase shift keying, the phase of carrier signal is shifted in accordance with the change in input bit stream. Whenever there is a change in data, i.e. from 0 to 1 or 1 to 0, the phase is shifted by 180° . However, the amplitude and frequency remain constant. This is also a binary PSK (BPSK). The PSK

is more resistant to interference compared to the FSK. Figure 1.10(b) shows the PSK representation of the example bit stream 1001.

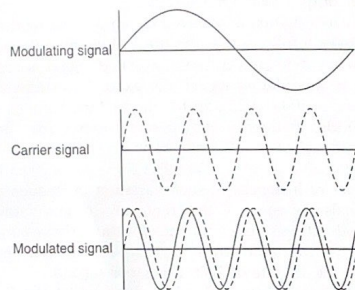


Figure 1.10(a) Phase modulation.

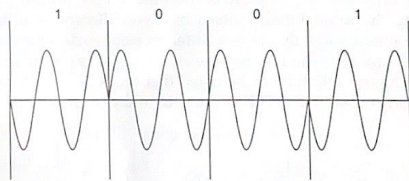


Figure 1.10(b) Phase shift keying.

1.2.5 Regulation on Usage of Radio Frequency

The spectrum (< 100 GHz) of the electromagnetic signal is prone to create interference with other services operating on the same frequencies. For example, short-range microwave transmissions operating on the same frequency must be kept separated by a reasonable distance. The avoidance of interference requires coordination between the users, and that eventually necessitates the regulatory boards or commissions.

There are agencies involved in setting the technical and maintenance standards, as well as laying down planning rules, performance targets and operational procedures for the management of telecommunication networks. Their target is to outline the recommendations of ITU (International Telecommunication Union) as the world's most authoritative body on telecommunications standards. One such organisation is International Frequency Registration Board (IFRB). It is the part of ITU

and is responsible for maintaining the list of radio frequencies used worldwide. It also conducts technical and planning studies for ITU.

In addition to the international bodies, some of the regional bodies such as FCC (Federal Communications Commission) and CEPT (Conference Europeenne des Postes et des Telecommunication) are responsible for controlling specific ranges of the spectrum in the US and Europe respectively. The following are the issues involved in regulation of radio frequencies:

- Amount of power transmitted
- Physical configuration of the equipment and antenna
- Issuance of license to the user
- Annual renewal of license
- Setting up of an acceptable transmission system
- Maintaining compliance

1.3 SUMMARY

Summarily, if we compare with wired network, the counterparts in wireless can be made as wireless LANs, wireless MANs (metropolitan area network) and wireless WANs (wide area network). Wireless LANs (e.g. range—spanning two adjacent buildings) are categorized further as the personal area networks (Bluetooth, data rate 1Mbps, range 10 metre) and Business LANs (802.11b, data rate 11 Mbps, range 100 metre). Examples of wireless MANs (e.g. range—spanning a city) is the wireless Local Loop (data rate 37 Mbps to 1.25 Gbps). On the other hand, wireless WANs (e.g. range—spanning earth) having wide coverage can further be categorized as cellular networks (GSM, data rate 9.6 kbps), Satellite systems (Motorola Iridium, data rate 64 Mbps) and Paging networks (ReFLEX, data rate 6.4 kbps).

The book is intended to address different issues in wireless networks. As mobile computing covers computing issues related to the voice and data communication over wireless network, we have chosen one representative network from each category of wireless network and elaborated those throughout the book. To start with, a sizeable number of chapters discuss cellular network, an example network of the WAN category, including its implementation standards from 2 G to 2.5 G. The next few chapters are dedicated to describe example networks of the other categories, namely MAN and LAN along with their standards and applications. The rest of the chapters concentrate on mobile data services including the issues of WAP (Wireless Application Protocol) and Mobile IP. Finally, an overview of 3G cellular networks (e.g. UMTS) is introduced in the last chapter.

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REVIEW QUESTIONS

1. Prepare a write-up describing evolution of mobile communication services.
2. Give an example of wireless system each from 1 G, 2 G, 2.5 G and 3 G.
3. Compare analog and digital signal.
4. Define wavelength and frequency of a signal. What is the relationship between these two parameters?
5. What is the relation between a signal's spectrum and its bandwidth?
6. How do you characterize an electromagnetic signal?
7. (a) What are the transmission ranges of the following signals?
 - (i) High Frequency (HF) signal
 - (ii) Super High Frequency (SHF) signal and Extremely High Frequency (EHF) signal
 (b) On which parameters transmission range of a signal depends?
8. What is information-carrying capacity of a signal? How is it related to bandwidth?
9. What is the concept of channel? Explain it with an example.
10. What is noise? How it affects signal transmission?
11. Define channel capacity and signal strength.
12. What is meant by $\text{SNR} = 20 \text{ dB}$?
13. (a) Define simplex, half-duplex and full-duplex communication. Give examples of each such communication systems.
 (b) How full-duplexing is implemented in wireless system?
14. What is the significance of 'multiple access'? Name important multiple access techniques.
15. Draw the block diagram of a radio system at both the transmitter and receiver sides. Give a brief description of the tasks of each of the modules in the diagram.
16. What is the need of modulation? Differentiate between modulating and modulated signal.
17. What are the two major tasks of modulation? Explain.
18. What are the principles of AM and ASK, FM and FSK and PM and PSK?
19. How binary values are represented in frequency shift keying? Explain it citing an example.
20. What are the merits of FM over AM? In which way PM is advantageous over FM?
21. (a) Why regulation is essential on the usage of radio spectrum?
 (b) Name an international agency/organization for monitoring of radio spectrum usage worldwide.
 (c) Name radio spectrum regulatory agencies in the US and Europe.
 (d) What are the issues of consideration under the purview of regulation?

2

WIRELESS WIDE AREA NETWORK (CELLULAR NETWORK)

Since long, people have sought to communicate on a worldwide basis with each other. It is no more a privileged service; rather it is a bare need. Worldwide communication has become possible due to technological advancements in diverse fields of telephony and the Internet. The dream of communication, even in geographically remote areas, has come true through the introduction of wireless networks.

The most popular way to categorize communication networks is based on their scale—that is, the magnitude of area coverage. The Wide Area Network (WAN) covers a large geographical area. The Internet, spanning the earth, is the most prominent example of WAN in the wired network domain. In the wireless domain, the major example of WAN is the cellular network. This chapter deals with the cellular network and investigates the internals of a wireless WAN.

2.1 THE CELLULAR CONCEPT

Traditionally, the WAN is a collection of Local Area Networks (LANs) dispersed geographically. The relatively new cellular network is the collection of small networks, operated by the service provider(s) local to the networks. This concept was introduced in the early 1970s at Bell Laboratories. One of the most successful initial implementations of cellular concept was the Advanced Mobile Phone System (AMPS). It has been popular in the United States since 1983.

(The system capacity of a cellular mobile network can be interpreted as the maximum number of users that can be supported at a particular point of time.) Support to a large number of users leads to a large area



coverage. If a single transmitter is used to cover a large geographical area, a very powerful transmitter/antenna has to be installed. However, in reality, the system capacity offered by a single high power transmitter with full set of radio frequencies, allocated to the system, can't extend beyond a limit. Therefore, the alternative is to use a set of radio frequencies to serve a comparatively smaller geographical area and then reuse it for serving the other small areas. It avoids installation of the high power transmitter while keeping the desired system capacity. Care must be taken to ensure that the same set of frequencies, used to serve more than one geographical area, does not introduce interference among signals from the users in two different areas.

Partitioning a large coverage area with the target to provide small contiguous areas, supported by a low power transmitter with low antenna height in each of the areas, is the basis of a cellular communication network. The implementation of such a system follows the concept of cellular radios.

The next subsection defines the components of a cellular mobile network.

2.1.1 The Components

The basic components of a cellular mobile network developed around the cellular concept are as follows.

Transceiver: The device capable of simultaneously transmitting and receiving radio signals is called a transceiver.

Mobile Station (MS): Mobile station is an electronic device used either as a portable hand-held unit or mounted in a vehicle for communicating voice and data. It contains a transceiver, antenna and control circuitry.

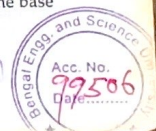
Base Station (BS): A base station is a fixed station in the mobile radio system and is used for radio communication with mobile station. It is normally located at the centre (Figure 2.1) of a network cell. Alternatively, it may also be placed on the edge of a coverage region. A base station contains transmitter, receiver, control unit, and antenna; and it has radio channels for allocation.

Mobile Switching Centre (MSC): An MSC coordinates routing of calls in a large service area. It connects both the base station and mobile stations to a telephone network called Public Switch Telephone Network (PSTN). That is, the MSC is an interface between the radio system and PSTN. The MSC is also referred to as the Mobile Telephone Switching Office (MTSO).

Forward Voice Channel (FVC) or downlink channel: The channels used for voice transmission from a base station to the mobile station are called FVC or downlink channel.

Reverse Voice Channel (RVC) or uplink channel: The RVCs or uplink channels are used for voice transmission from a mobile station to the base station.

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Forward Control Channel (FCC)/Reverse Control Channel (RCC): These two channels (FCC and RCC) are responsible for initiation of mobile calls and are often called set-up channels. The FCC continuously broadcasts all traffic requests targeted to the MSs and serves as beacon. In general, on an average 5% of the total available channels within a system are assigned as the control or set-up channels.

2.1.2 Cellular Architecture

Figure 2.1 illustrates the cellular network architecture with its components, the MS, BS and MSC. The whole geographical area covered by the network is partitioned into a number of hexagonal cells. Each such cell is serviced by a BS and a set of BSs are connected to an MSC. In addition to its function as a switching exchange, an MSC handles/controls (mobility management) of MSs in coordination with the BSs. The MSCs of the network are connected to the PSTN through a wired connection. The only wireless connection in a cellular network is in between an (MS and the BSs) — *wireless*.

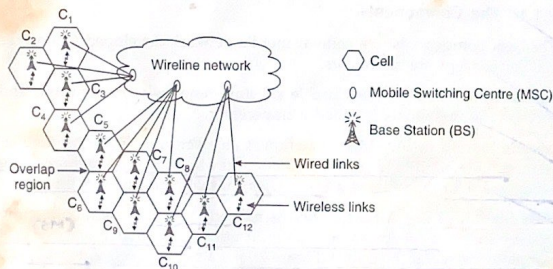


Figure 2.1 A cellular mobile network architecture.

The hexagonal cell shape considered in Figure 2.1 is purely imaginary. In reality, no such cell shape exists in the cellular network implementation. However, for proper planning of the system architecture, the hexagonal shape is universally adopted for several reasons.

Cell shape: In a cellular network, the geographical coverage region is considered to be partitioned into a number of small areas called cells. These cells are hexagonal in shape. The actual radio coverage of an antenna installed within a network cell is called the footprint. Although the real footprint is amorphous (the assumption of regular cell shape, in cellular network, is essential for a systematic system design) The probable candidate regular shapes can be the square, circular, hexagonal, etc. In a square cell based design, all the adjacent cell centres cannot be at the

same distance. The normal practice of installing BS antenna at the centre of a cell, therefore, cannot ensure the equidistance among the antennas. The antennas placed at equidistant simplify the task of determining when to switch a user (MS) to an adjacent antenna and as well as to choose the right one.

Considering circular radiation pattern and free-space propagation of the omni-directional BS antenna, the circular cell shape is mostly desirable. However, the circular cell can't fully cover a geographical region without the overlapping areas. On the contrary, a hexagonal cell approximates the circular radiation pattern. The adjacent BS antennas placed at the centres of such hexagonal cells are equidistant. Moreover, for a given distance between the centre of a polygon (circle/square/hexagon) and the farthest points on its perimeter, the hexagon has the largest area coverage among the three shapes.

The above discussions point to the fact that a hexagonal cell shape is most desirable in cellular network to cover a given network service area (coverage area). However, the exact hexagonal shape of cells cannot be implemented due to topographical and some other practical limitations of installing base station at a desired site. In reality, the designers practice variations in cell shapes. The reasons for such variations are:

- Due to practical limitations, e.g. land dispute, location unsuitability, some of the BSs can't be placed at the desired point (centre of a hexagon).
- The antenna at BS has some directional bias, in spite of the fact that it is desired to be omni-directional.
- The signal propagation characteristics may vary at places. In some cases, the transmission path between a transmitter and the receiver can be the simple direct line of sight. In a number of situations, the transmission path may be obstructed severely by high-rise buildings, foliage and terrain.

2.2 CALL SET-UP

When a Mobile Station (MS) is turned on within the coverage of a network, the following tasks are executed:

- (i) The MS scans the group of FCCs (Forward Control Channels) in search of the strongest BS signal. The BS from which the strongest signal is received works as serving BS of the call initiator MS.
- (ii) It checks continuously the selected FCC to ensure that the signal is not below the usable level.

In a cellular network, the call may be initiated by an MS or by a land phone. The call initiation process followed by an MS is different from that of a land phone. The steps to be executed during a call initiation are summarized in the next two subsections.

2.2.1 Call Initiation by an MS

Despite variations in the characteristics of MSs, the following sequential steps are commonly performed while a call is initiated by an MS:

- (i) A call request is sent from the MS to the serving BS on RCC (Reverse Control Channel). The request message contains the MIN (Mobile Identification Number), ESN (Electronic Serial Number) and phone number of the called party (MS/land phone).
- (ii) On receiving the request message, the serving BS transfers the request to the concerned MSC (Mobile Switching Centre).
- (iii) The MSC, on receiving the message, validates the request/MS.
- (iv) After successful validation, the MSC instructs the BS to assign a pair of unused FVC and RVC to the caller MS.
- (v) MSC connects the MS with the called party. If the called party is also an MS,
 - (a) MSC dispatches the request to all its BSs.
 - (b) The MIN of the called MS is paged by all the BSs and instructs the called MS to use the unassigned FVC and RVC pair.
 - (c) The called MS responds over RCC.
 - (d) The serving BS sends the acknowledgement to MSC.
 - (e) The serving BS sends an alert message to the called MS to ring.
- (vi) Both the caller and called MSs begin voice transmission and reception.

2.2.2 Call Initiation by a Land Phone

The call initiation from a land phone leads to the following steps of activities while connecting an MS.

- (i) The MSC receives the call request from a land phone through PSTN.
- (ii) The MSC connects the land phone with the called party (MS)
 - (a) MSC dispatches the call request to all its BSs.
 - (b) On receiving the call request message the MIN of called MS is broadcast by all the BSs.
 - (c) The called MS responds over RCC.
 - (d) The serving BS sends an acknowledgement to the MSC.
 - (e) The serving BS sends an alert message to the called MS to ring.
- (iii) On receiving the acknowledgement, MSC instructs the BS to assign a pair of unused FVC and RVC to the called MS.
- (iv) The MS responds to the caller party (land phone) over RVC for voice transmission.

2.3 FREQUENCY REUSE AND CO-CHANNEL CELL

In a cellular mobile network, the system may have to support a large number of simultaneous calls from and to the MSs. It requires allocation of a large number of forward and reverse channel pairs, thereby requiring a wide frequency bandwidth, which is a scarce resource. The objective of frequency reuse in a cellular system is to employ the same frequency in more than one network cell. This reuse reduces the demand of such scarce resource. However, use of the same frequency in nearby cells may interfere preventing the desired information flow. Therefore, the design challenge is to determine the distance between two cells that can use the same set of frequencies within the acceptable limit of interference.

The frequency reuse ensures proper distribution of total available frequencies allocated to a network. The total allocated frequencies are partitioned into a number of distinct sets for distribution in different network cells. A group of adjacent cells, called the cluster or compact pattern, is then determined. Each cell of a cluster is then allocated a distinct set of frequencies. It implies that in a cluster/compact pattern there is no scope of frequency reuse. A frequency f is allocated to two different cells X_i and X_j , ensures that X_i and X_j belong to two different compact patterns. Co-channel cell: The cells that can use the same set of frequencies as assigned to a cell X are called the co-channel cells of X . In Figure 2.2, the set of cells $S = \{F_1, F_2, \dots, F_{19}\}$ form the cluster/compact pattern. Each cell of S uses a unique set of frequencies from the total allocated frequencies

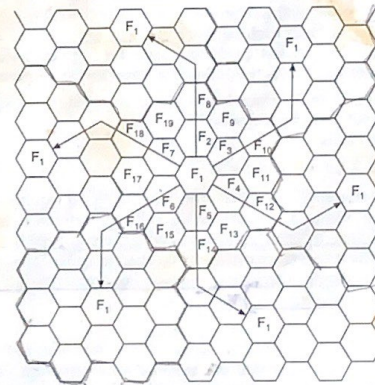


Figure 2.2 The co-channel cells.

of the network. The cells labeled with F_1 and are located outside the compact pattern are the nearest co-channel cells of the cell that is labeled with F_1 and is located at the centre of the compact pattern. The co-channel cells use the same set of frequencies for allocation to the MSs under their control. The minimum distance between the centres of any two co-channel cells that are nearest to each other is D . The distance D defines the effective measure of frequency reuse while avoiding any possibility of interference beyond the acceptable limit.

The repetition of compact pattern in the network coverage area places the co-channel cells in tiers. In general, a cell is surrounded by 6 T co-channel cells in tier T . Figure 2.2 indicates 6 co-channel cells F_1 only at tier 1 ($T = 1$)—that is, the nearest co-channel cells. Similarly, tier 2 contains 12 co-channel cells. The co-channel cells of a network cell in each tier form a hexagon. The minimum distance between a cell (centre cell labeled with F_1) and its co-channel cell at tier T is TD .

The number of cells in a cluster $N = i^2 + ij + j^2$, where i and j are two positive integers. The size of cluster depends on the parameters i and j . In the figure $i = 3$, $j = 2$ and, therefore $N = 19$. The N is also considered as the measure of frequency reuse factor. In TDMA and FDMA, the frequency reuse factor is greater than 1 whereas in CDMA, this reuse factor is 1. In some literature $1/N$ is considered as reuse factor.

Example 2.1 Let us consider for a cellular system, 33 MHz bandwidth is allocated. Each of the full-duplex voice and control channels uses 50 kHz bandwidth. The number of channels can be allocated per cluster is, therefore, $33000/50 = 660$. Now, if the cluster size chosen is $N = 12$ ($i = 2$, $j = 2$), for a cluster of size 12, then each network cell gets $660/12 = 55$ channels.

To identify the nearest co-channel cells for a given cell X , and for particular values of i and j , the following steps are to be performed:

- Starting from X , move i cells along any chain of hexagons
- Turn 60° counterclockwise
- Move j cells

In Figure 2.2, $i = 3$ and $j = 2$.

Co-channel interference: A signal other than the signal required for desired communication within a network cell is called interference. Interference creates disturbance in communication and affects adversely the system performance. The interference caused by co-channel cell is called co-channel interference. Co-channel interference plays an important role in determining the quality of service (QoS).¹

¹ The QoS is measured by several parameters such as coverage, call blocking (unable to initiate a call), call dropping (call is stopped during conversation), audio quality (cross-talk), etc. The audio quality is mainly affected by co-channel interference.

conflicting $\begin{cases} \text{system capacity} \\ \text{quality of service} \end{cases}$

2.4 CELL DESIGN

The allocated frequency spectrum in a cellular network is very limited. The maximization of system capacity, keeping a reasonable quality of service, under the constraint of limited spectrum is the measure of performance of a system. However, the system capacity and quality of service are the two conflicting requirements. Thus while selecting the cell and cluster size, these two constraints are to be considered.

For example, let us assume that a cellular system has S duplex channels and a group of K channels are allocated to each cell of the network. If N cells are in a cluster, then $S = KN$. If a cluster is replicated M times, the total number of allocated duplex channels is $C = MKN = MS$, out of which S channels are unique. Here C denotes the capacity of the network. It signifies that the capacity is directly proportional to the number of times a cluster is replicated. Figure 2.3 shows the two parameters D and R (R is radius of a cell) for $N = 7$.

$$C \propto M$$

$$C = MKN$$

$$= MS$$

S channels are unique.

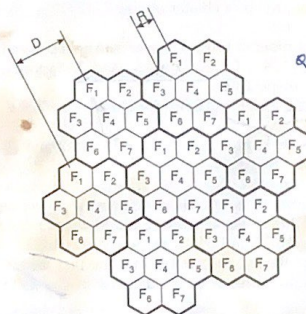


Figure 2.3 7-cell reuse pattern.

The ratio $Q = D/R$ is referred to as the frequency reuse ratio. From hexagonal cell geometry, the relation between frequency reuse ratio (Q) and the number of network cells per cluster (N) is

$$Q = \frac{D}{R} = \sqrt{3N} \quad (2.1)$$

The number of cells in a cluster (cluster size) is $N \propto D/R$. Now for a fixed cell size (R), if the cluster size (N) is increased, then D is to be increased. Alternatively, a large cluster size indicates that D/R is large. On the other hand, a network with small cluster size indicates its co-channel cells are located closer to each other and hence co-channel interference is significant.

The above discussions set the design criteria of a cellular network for the given coverage. These are (i) to maximize capacity ($C = M \times K \times N$), the smallest possible value of N is to be chosen and (ii) to reduce interference, the value of N —that is, the cluster size, is to be large. Thus to meet these two conflicting requirements (i) and (ii), a trade-off between the capacity and interference is desirable. (v. imp.)

Example 2.2 Consider a cellular system having 1596 duplex channels to cover 2310 sq. km. The area of each cell is 6 sq. km. Assume that we need to compute the following:

- (i) System capacity (C) for cluster size 7, 12 and 19. $C = M \times K \times N$
 (ii) M refers to number of times the cluster is replicated to cover the entire network area.

The total allocated channels in the system is $S = 1596$.

For the cluster size $N = 7, 12$ and 19 , the number of channels per cell (K) = $1596/7, 1596/12$ and $1596/19 = 228, 133$ and 84 respectively.

The coverage area of a cluster of size $7/12/19$ is $7 \times 6/12 \times 6/19 \times 6 = 42/72/114$ sq. km.

That is, the number of times a cluster is to be replicated is $M = 2310/42, 2310/72$ and $2310/114 = 55, 32$ (approx) and 20 (approx) for the cluster size 7, 12 and 19 respectively.

Therefore, the system capacity for cluster size 7 is $C = M \times K \times N = 55 \times 228 \times 7 = 87780$.

For cluster size 12 and 19, $C = 32 \times 133 \times 12$ and $20 \times 84 \times 19$, that is, 51072 and 31920 respectively.

The system capacity $C = 87780$ signifies that at any point of time maximum 87780 simultaneous calls are allowed in the network with cluster size 7. Similarly, if the cluster sizes are 12 and 19, the number of simultaneous calls permitted is 51072 and 31920 respectively.

The results of the above example indicate that the system capacity of a cellular system decreases with the increase in cluster size.

2.5 INTERFERENCE IN CELLULAR SYSTEM

Interference creates disturbance in communication and affects the performance of a cellular network. It causes impairments ranging from the decrease in system capacity to dropping of calls. Signal received by the base station (BS) from a communicating mobile station (MS) may encounter interference due to:

- transmission from other mobiles in the same cell
- transmission from other mobiles in the neighbouring cells
- background noise
- other neighbouring base stations operating in the same frequency band
- noncellular system leaks energy into the cellular frequency band

Some of the reasons for interference are the improper channel assignment strategies, co-channel interference and adjacent channel interference out of imperfect cell design, etc. The co-channel interference is described in Section 2.3.

Adjacent channel interference: Signal impairment to a frequency due to the presence of another signal of very close frequency is called 'adjacent channel interference'. It mainly occurs due to the imperfect receiver filters that allow close/adjacent frequencies to leak into the passband².

The adjacent channel interference can have a serious effect if two MSs (adjacent channel users), one is located very close to the BS and the other is at a distance from the BS, simultaneously transmit equal power. The BS receives more power from the MS close to it. The SNR (signal-to-noise ratio) for this communication is much higher than that for the MS at a distance. Such a case is referred to as the 'near-far' problem. Further, if the MS close to the BS transmits a signal with order of magnitude much higher than the MS at a distance, the SNR for the second MS may not be detectable. This effectively creates jams in the communication channel of a network.

2.5.1 Signal-to-Interference Ratio

The quality of a signal communicated is also measured by a parameter called (S/I) (signal-to-interference) ratio. To compute S/I , the following interference sources are ignored as a matter of convention:

- co-channel interference from the second and other higher tiers of the network
- adjacent channel interference (negligible in comparison to the co-channel interference)
- background noise

The inter cell interference is mainly dominated by the co-channel interference.

The average channel quality is a function of distance dependent path loss³. Effects of other issues of communication are normally ignored. If c be the number of co-channel interfering cells and I_i be the interference

² In digital communication, the frequency band is split up into two main parts—the baseband and the passband. The passband contains all frequencies above a limiting frequency whereas the baseband refers to the frequencies below the limiting frequency. In radio communication, a baseband signal must be converted to the passband signal for its successful transmission.

³ Path loss is the reduction in power density of the radio signal as it propagates. It is one of the dominating elements that causes impairment in the propagation channel as well as distorts the information-carrying signal as it propagates over medium. As an MS moves away from its serving BS, the received signal at BS becomes weaker because of the reduction of power density of the signal due to the growing distance between the MS and the BS.

power caused by transmissions from i th interfering co-channel cell to the BS, the S/I at the desired mobile receiver is:

$$\frac{S}{I} = \frac{S}{\sum_{i=1}^C I_i} \quad (2.2)$$

Now, if r be the distance between the mobile station and serving BS, the desired received signal power S is proportional to r^{-k} , where k is the path loss component (in general, $2 \leq k \leq 5$). Further, if D_i is the distance between i th co-channel cell and the mobile station, the I_i is proportional to $(D_i)^{-k}$. Thus the received S/I at the mobile station can be expressed as:

$$\frac{S}{I} = \frac{r^{-k}}{\sum_{i=1}^C (D_i)^{-k}} \quad (2.3)$$

If only first-tier co-channel cells are considered, then $C = 6$, and when an MS is in cell boundary ($r = R$), then $D_i = D$, for $i = 1, \dots, 6$. Therefore, from Eq. (2.3), it can be deduced that:

$$\frac{S}{I} = \frac{\left(\frac{D}{R}\right)^k}{6} = \frac{Q^k}{6} = \frac{(\sqrt{3N})^k}{6}$$

or,

$$Q = \left(6 \times \frac{S}{I}\right)^{1/k} \quad (2.4)$$

It implies, from Eq. (2.4), for a given acceptable signal-to-interference (S/I) ratio, the frequency reuse ratio Q can be computed. From a known Q , the cluster size/compact pattern can also be settled (Eq. (2.1)) for a network.

Example 2.3 Let us assume that for a GSM TDMA system, the acceptable S/I is 15 dB or more. If path loss exponent $k = 3$, the frequency reuse ratio $Q = (6 \times 10^{1.5})^{1/3} = 3.13$. Then the compact pattern size $N = Q^2/3 = 3.26 \approx 3$. That is, a 3-cell-frequency reuse system is to be used with 15 dB S/I .

2.5.2 Interference Reduction

It is reported in the earlier section that the co-channel interference is reduced if the distance between co-channel cells is increased. However, the large distance among the co-channel cells increases the cluster size, thereby reduces the system capacity.

On the other hand, the adjacent channel interference can be reduced through careful filtering and the proper channel assignments. Since each cell of a network is given only a fraction of the total available channels, a cell need not be assigned channels that are adjacent in frequency. This enables provision for the greatest possible frequency separation between the channels to ensure reduced adjacent channel interference within a cell.

Controlling transmission power of mobile stations (MSs) is another effective option to reduce the overall signal-to-noise ratio caused due to reverse channel transmission. In such mechanism, the power levels transmitted by each mobile station are under constant control of the serving base station (BS) to ensure that each unit transmits the minimum power necessary to maintain a good quality link on the reverse channel. The transmission power control also helps to extend battery life and combats the 'near-far' problem. (The MS closer to BS transmits less power compared to that of the MS at a distance while both the MSs are adjacent channel users. At the serving BS's receiver, it maintains almost the same SNR for all the MSs' transmitters.)

2.6 CHANNEL ASSIGNMENT

At the time of installation of a cellular network, each cell is assigned a set of channels to provide services to the individual calls of the cell. (The task of assigning frequency channels satisfying the frequency separation constraints is known as the channel assignment problem.) As the use of mobile communication systems grows, more efficient channel assignment/allocation techniques are needed. The channels on a network are the scarce resources and its proper management is very much desirable. An efficient assignment is not only important for new call initiation but it has also a great impact on call management while a mobile station moves from a cell to another—that is, handoff.

Channel assignment strategies are classified as the fixed and dynamic assignment. In fixed assignment, each cell of the network is to be allocated a predetermined set of voice channels. To provide call service, the cell can only utilize channels from that set. Any new call attempt within the cell is served while considering the channels remain unused at that moment. If all the channels of that cell are occupied, the attempted call is blocked.

One of the variations of fixed channel assignment is the channel borrowing. In this strategy, a cell is allowed to borrow channels from a neighbouring cell when its channels are already occupied and a handoff/new call is to be serviced. The MSC supervises such borrowing processes ensuring that this does not disrupt or interfere with the calls in progress. That is, the channels currently not being used by the donor cell or by its co-channel cells can be borrowed.

In dynamic assignment, the voice channels allocation to cells are not predetermined. Each time a call request is made by an MS, the serving BS seeks a channel from the MSC. The MSC in turn allocates a channel to the cell, requesting for the channel. While assigning channels, an MSC should consider the following:

- Likelihood of future blocking within the cell
- Reuse distance
- Cost functions

A number of works are so far been reported on channel assignment strategies. The main objective of such works is to ensure maximum utilization of radio resource, keeping the call blocking probability as low as possible. Such a region/area is defined as the hotspot. Therefore, an effective channel assignment strategy has to take care of the hot-spot areas.

2.7 HANDOFF

During its move an MS (mobile station) may occasionally switch to a new cell of the network. Switching from a cell to its neighbouring cell implies change of BS (base station) as well as switching to new voice and control channels. The process of transferring an MS from the control of current serving BS to a new one is called handoff. That is, the handoff process includes identification of a new BS for the MS and also allocation of the voice and control channels associated with the new BS. Handoff must be performed successfully and as infrequently as possible to reduce overhead due to switching. *v. Imp*

The early prediction of handoff is the requirement for an efficient handoff-handling scheme. The simplest implementation is that whenever the power level from an MS received by a BS crosses a threshold ($P_{\text{threshold}}$), the BS initiates the handoff process. However, the whole process should be transparent to the users. To realize it, the system designer must specify an optimal signal level ($P_{\text{threshold}}$) to initiate a handoff. This threshold signal level is defined as $P_{\text{threshold}} = P_{\text{minimum}} + \Delta$, where P_{minimum} is the minimum usable signal for acceptable voice quality at BS receiver and Δ is a small value.

If $P_{\text{threshold}}$ is too large, there may be unnecessary handoffs. The unnecessary handoff is considered to be costly to an MSC. On the contrary, if $P_{\text{threshold}}$ is too small, the system may not get sufficient time to complete the handoff process. While the BS/MSC handling the handoff, the MS may move farther from the serving BS and the call may be lost due to weak signal. As the handoff is costly, prior to any handoff, the system has to ensure that the drop in the measured signal level at serving BS is not due to any other reason except the movement of the MS away from the serving BS. In reality, a serving BS monitors the signal level received from an MS for a certain period of time. If the MS moves very fast, the slope of the short-term average of received signal becomes steep indicating that the handoff is to be processed instantly. *v. Imp*

2.7.1 Handoff Strategies

The handoff in a cellular network is a regular feature. A cellular network without handoff is of no practical use. Therefore, efficient strategies need to be devised to address this issue. The following handoff tackling strategies are considered to be effective in a cellular mobile network:

- Network-controlled handoff (NCHO)
- Mobile-assisted handoff (MAHO)

NCHO: In this technique, the BSs monitor the signal quality received from an MS and report it to the MSC. On receiving this information from the BSs, the MSC selects the new serving BS and initiates handoff. The new BS takes control of the MS.

MAHO: In mobile-assisted handoff, implemented in GSM, the process of handoff is jointly taken care of by the mobile station (MS) and the network. The MS measures the signal levels received by it from the various BSs using a periodic beacon, generated by the BSs, to keep track of the MSs. The MS then feeds the measured signal levels back to the MSC via the serving BS. The MSC finally takes the decision of handoff.

Based on the procedures followed to take care of the handoff issues in a cellular network, the handoff is classified as Hard and Soft handoff.

Hard handoff: In Time Division Multiple Access (TDMA) and Frequency Division Multiple Access (FDMA), the frequency reuse factor (N) is greater than one. The neighbouring cells within a cluster use different frequency bands of the available frequency spectrum in such an implementation. An MS in a cell X' receives signals from different cells (BSs) surrounding X' . When the signal strength received by an MS from a neighbouring cell (X) exceeds a predefined threshold, the MS is instructed to switch to it (X). A new frequency band is then allocated to the cell X to support the ongoing communication of the MS. Such a termination of the existing connection (with X') to establish a new connection (to X) is referred to as the Hard Handoff. In hard handoff, an MS can have the radio link only with a single base station at any point of time and, therefore, it is also known as the 'break-before-make'. *v. Imp*

Soft handoff: In soft handoff, the new connection between an MS and the new BS is established prior to the termination of existing link between the MS and the currently serving BS. Unlike hard handoff, during soft handoff, an MS can simultaneously communicate to more than one BS. The Code Division Multiple Access (CDMA) systems implement soft handoff techniques. As in CDMA systems for the cluster size $N = 1$ (Section 2.3), the spatial separation of frequency spectrum for allocating separate frequency band in cells is not necessary. Further, the interference from neighbouring cells is not too serious in such systems. Therefore, it can allow an MS to be connected to more than one BS simultaneously. If the MS enters a region in which the transmissions from two BSs are comparable, the MS then gets connected to the two BSs. The MS remains connected to the two BSs until one BS clearly predominates and the MS is assigned to that cell. *v. Imp*

1st generation vs 2nd generation handoff

In 1st generation cellular systems, the voice transmission is analog. It uses FDMA technique to share the available spectrum. As cellular mobile systems have grown from analog to digital, the techniques of handling handoff get

the maturity. In 1st generation analog cellular systems, the notable features of handoff processing techniques are:

- Handoff decisions are taken by MSC with assistance of BS
 - ♦ BS measures signal strength (SS)
 - ♦ MSC supervises
- Each BS constantly monitors SS of RVCs (reverse voice channels) and determines the relative location of each MS

On the other hand, in 2nd generation digital cellular systems, which use TDMA technology, the following are the important features for implementing handoff:

- Handoff decisions are mobile assisted (MAHO)
- MS measures the power levels received from the BSs of neighbouring cells
- The measured values are continually sent to the serving BS

In MAHO, handover of calls between the BSs is much faster than that realized in 1st generation systems. An MS continuously measures the parameter values required for the handoff. The MSC no longer remains constantly busy for monitoring the signal strength at the MS. MAHO is particularly suitable for micro cellular (to be discussed in Section 2.7.2) environments where handoffs are more frequent. (v. Imp)

Prioritizing handoffs

From the user's point of view, abrupt termination of a connection/call during running conversation is more annoying than a call is blocked occasionally on a new call attempt. (Hence prioritizing handoff over the new call attempt is desirable.) That is, at an instant of time if the system has to support a handoff and a new call attempt simultaneously, then a strategy is to be adopted so that service to handoff is considered first. There are two standard handoff strategies—the guard channel concept and the queuing of handoff requests. (i)

In the guard channel, a fraction of the total available channels within a network cell is reserved for providing service to the handoff requests from ongoing calls. It means a cell gives priority to the handoff calls. On the other hand, in the queuing based handoff handling technique, there is no such reserved channel for the handoff calls. At the time of handoff if there is no available channel to serve, the call request is queued. Therefore, it decreases probability of forced termination of a call but delays the handoff.

The handoff strategies, guard channel concept and queuing of handoff, limit the total carried traffic in a network. In guard channel based technique a comparatively fewer number of channels can be allocated for the new call attempts by the MSs as a set of channels is kept aside for handoff calls. This reduces the total carried traffic within the system at any instant of time. On the other hand, in queuing the probability of a successful handoff improves at the cost of increased blocking probability of originating calls.

2.7.2 Constraints

The practical constraints in performing handoff are the accommodating users (MSs) with different speeds and the cell dragging.

Users with different speeds

In the cellular network, the mobile users can be inside a high-speed vehicle or they can be simply pedestrians. For pedestrians carrying MSs, the handoffs do not take place very frequently. So, the cell size in the network can be kept smaller (micro cell approach) without any severe problem in processing handoff. On the other hand, for fast moving users (sitting in a high speed vehicle) micro cell approach is not desirable. Micro cell based design for such cases may result in frequent events of handoff, thereby, increases computational overhead at the system components e.g. BS, MSC, etc. For high-speed mobile stations, the cell size is to be large (macro cell approach) to ensure efficient handling of handoff avoiding the excess loads in system components.

Theoretically, the cellular concept provides additional system capacity through addition of new cell sites. However, in reality, it is difficult for the service providers to identify new cell sites in urban areas even for some non-technical reasons. The practical solution is to use different antenna heights as well as the different power levels in the same location, to enable the design of 'micro' and 'macro' cells. This technique is called the 'umbrella cell' approach (Figure 2.4). The umbrella cell configuration consists of one large BS (macro cell) with high transmission power and antenna height. This set-up serves as an 'umbrella' for a number of small BSs (cells) with low transmission power and small diameters. While a high-speed user

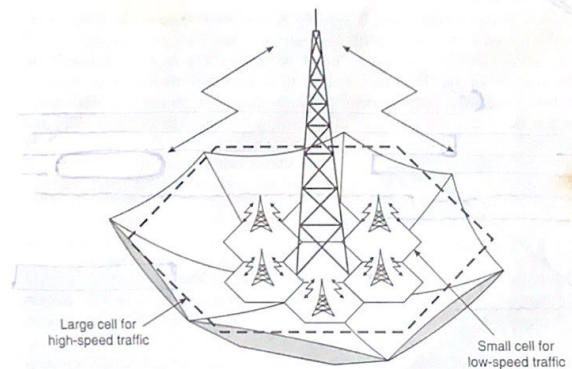


Figure 2.4 The umbrella cell.

(MS) within the large umbrella cell approaches the large BS, with rapidly decreasing speed, the BS decides to handover the MS into one of the co-located small BSs (micro cells) without any intervention of the MSC.

In GSM, the speed of a user can be determined almost accurately by the change of value of the parameter, timing advance (TA)⁴. Its value is updated in the base station controller (BSC) located between BS and the MSC every 480 ms.

The umbrella cell based system (service area is covered by co-located micro and macro cells) has the following advantages over a microcell (service area is covered by microcells only) based system:

- Since it is not necessary to cover the whole service area with microcells, infrastructure cost is saved.
- The number of handoffs is much less than in microcell based system. In such a system whenever an MS, irrespective of its speed, crosses a cell boundary, a handoff occurs. On the contrary, in an umbrella cell environment, MSs with higher speed are under control of the BS of macrocell and the MSs with lower speed are under the control of the BS of microcell. So the MSs under macrocell are subject to less number of handoffs in comparison to the MSs under a microcell. Total number of handoffs is reduced in this way.

Frequent handoffs result in a substantial increase of the signalling load for a network. It degrades the quality of signal at the end user. The umbrella cell based system eliminates the possibility of frequent handoff for high speed MSs and very much effective in urban environments that feature city highways.

Cell dragging

In microcell environment, a serving base station may sense very strong signal from a slow-speed MS (pedestrian) even the user carrying the MS is well beyond the designated range of the cell. The signal received at the BS may be higher than the handoff threshold and, therefore, no handoff takes place. This phenomenon typically occurs in an urban environment when there is a line-of-sight (LOS) radio path between the user (MS) and the serving BS. It also causes interference when the MS penetrates deep into an adjacent cell area. Such a state is called the cell dragging. The parameter $P_{\text{threshold}}$ for taking handoff decision is to be chosen carefully to get rid of this problem.

2.7.3 Roaming

The roaming is a mechanism by means of which the intersystem handoff is taken care of. While the signal strength received by a mobile station (MS) becomes weak and the mobile switching centre (MSC) cannot find

⁴ In GSM, timing advance (TA) corresponds to the time a signal from the MS takes to reach to the base station.

a cell within its system to handover the MS, it is handed off to a cell under the control of a different MSC. The intersystem handoff is called roaming (Figure 2.5).

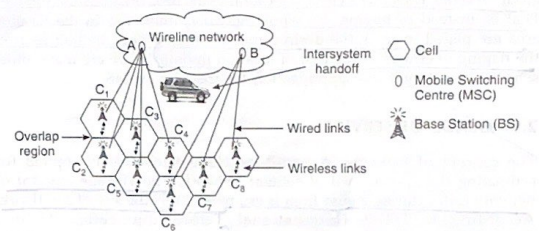


Figure 2.5 Intersystem handoff.

The important features of roaming are

- The MS moves out of its home network (to be described in Section 2.8).
- Local call becomes long distance call.
- Compatibility between two MSCs must be determined before implementing an intersystem handoff.

2.8 MOBILITY MANAGEMENT

The handoff management schemes are described in the earlier section. It is a part of mobility management in a cellular mobile network. The other important issue in mobility management is the location management. In this section, we concentrate on location management of a mobile station (MS) to keep track of its location for establishing seamless connection.

To start with an MS is assigned to a network called the home network of the MS. The association between the MS and the home network is made through a registration process. The home network is always aware of the current location of MS. While roaming from its home to a different network, an MS must register at the visiting network. Simultaneously, this update must reach its home network through the visiting network establishing an association between the home and the visiting networks.

A cellular network maintains two databases—Home Location Register (HLR) and Visiting Location Register (VLR). The HLR keeps the profile of mobile station and authenticates the subscriber before any updates. The VLR, on the other hand, authenticates the visiting MS in consultation with the HLR. The whole process which keeps track of a mobile subscriber's current location is called location management.

handoff management comes under mobility management

When a call/message destined to an MS is generated, the whole network is paged by the MSC to track the MS. In an alternative implementation, every time a mobile station crosses a location area (a group of cells), it sends its current location to the network (location update). That is, instead of paging the whole network, only cells in the location area are paged to track the desired mobile station. This method reduces the paging overhead at the cost of location update. There are many other schemes that target cost optimization for tracking an MS.

2.9 GRADE OF SERVICE

The concept of trunking in circuit-switching telephone is applied for estimating the system load of cellular network. The average amount of network traffic during a busy hour is estimated to fix the size of a network. According to ITU-T (International Telecommunication Union-Standardization Sector) recommendation, the busy hour is determined by considering the average of the busy hour traffic on the 30 busiest days of a year. Statistically, not all subscribers make calls at the same time and, therefore, it is reasonable to fix the size of network that can be able to handle expected level of load in the network. As a result of which whenever traffic load crosses the expected load, the probability of failure of an attempted call becomes non-zero and this may prohibit a user to access the trunked system. The ensured user access to a trunked system during the busiest hour is measured by the Grade of Service (GOS).

Cellular radio systems rely on trunking to accommodate a large number of users in a limited radio spectrum. There is a trade-off between the number of available pair of channels and the likelihood of a particular user finding that no channels are available during the peak time. C is the number of trunked channels offered (channel capacity) by a trunked radio system and A is the total offered traffic (offered load).

The trunking theory was developed by Danish mathematician Erlang. After his name a unit (Erlang) to measure the traffic intensity in a network is introduced. One Erlang represents the amount of traffic intensity carried by a channel when completely occupied. The traffic intensity A Erlang is expressed as $A = \gamma \times h$, where γ is the mean rate of connection requests attempted per unit time and h is the mean call holding time per successful connection. Thus A is the average number of calls arriving during the average holding period.

The traffic intensity A Erlang can also be expressed as per user traffic intensity (A_{pu}) and the number of users (n), i.e. $A = n \times A_{pu}$. If the average rate of connection request in a system is 25 calls/minute, and the average call holding time is 3 minutes, then the traffic intensity $A = 75$ (as $A = \gamma \times h$). This indicates if the channel capacity is exactly 75, it can meet the average demand. To meet the worst-case traffic load, the chosen capacity should be greater than 75. On the other hand, the channel capacity 150 indicates the channels in the system are remaining half-utilised.

In the trunked cellular system, when a user tries to communicate and all the channels are busy, then either the user is blocked, i.e. denied access to the system, or the user is queued until a channel is available. In the first case, the trunking is called blocked call cleared or lost call cleared (LCC); whereas the queuing is referred to as blocked call delayed or lost calls delayed (LCD). In general the cellular systems implement LCC. This model assumes the following:

- fixed arrival rate that follows poisson distribution
- memory-less arrivals of call requests, i.e. all users including blocked users may request a channel
- probability of a user occupying a channel is exponentially distributed so that long-duration calls are less likely to occur
- finite number of channels
- infinite number of users

An LCC model based system follows Erlang B formula to determine the blocking probability (P) of a new attempted call, where

$$P = \frac{A^C}{C!} \sum_{x=0}^C \frac{A^x}{x!}$$

is the measure of GOS. For example, GOS = 0.001 implies that during a busy hour, on an average, one out of 1000 call requests may be blocked. Table 2.1 presents sample values of offered load to achieve the desired GOS for different number of channels.

Table 2.1 Capacity of an Erlang B System

Number of channels (C)	Offered load				
	GOS = 0.001	GOS = 0.002	GOS = 0.005	GOS = 0.01	GOS = 0.02
10	3.09	3.43	3.96	4.46	5.08
20	9.41	10.07	11.10	12.03	13.18
30	16.68	17.61	19.03	20.34	21.93
40	24.44	25.60	27.38	29.01	30.99
50	32.51	33.88	35.98	37.90	40.25
60	40.79	42.35	44.76	46.95	49.64
70	49.24	50.98	53.66	56.11	59.13
80	57.81	59.72	62.67	65.36	68.69
90	66.48	68.56	71.75	74.68	78.30
100	75.24	77.47	80.91	84.06	87.97

The table enables computation of—

- the GOS (P) for a given offered load (A) and the number of channels (C).

- the traffic load (A) that can be handled for a given number of channels (C) to achieve a desired GOS (P).

Example 2.4 Let us consider the network with $C = 20$ (number of channels/cell) and on an average each user makes 3 calls/hr. The average duration of a call is 2 minutes. The number of users supported in a cell with 1% blocking ($GOS = 0.01$) can be estimated from Table 2.1.

Now, as $C = 20$ and $P = 0.01$, then from Table 2.1, $A = 12.03$. Moreover, $A_{pu} = \lambda \times h = 3 \times (2/60) = 0.1$ Erlang. So, the number of users admissible to a cell = $12.03/0.1 = 120$.

Example 2.5 In a system with $C = 120$ channels, if the utilization of channel is $2/3$, then the traffic intensity (load) in this network is $A = 2/3 \times 120 = 80$.

2.10 CAPACITY IMPROVING METHODS

The overall system capacity in a cellular network increases with the introduction of frequency reuse (Section 2.3). However, the frequency reuse cannot always avoid the cases of hotspot in a network. The channels assigned to the network cells may not be sufficient to accommodate all the users seeking connection. A set of cells may get over-committed resulting the hotspots (cells) within the network. This problem is taken care of by the technique called 'load balancing'.

To address the issue of hotspot, the capacity of a cellular mobile network can be increased simply by allocating/adding new channels. Alternatively, cells in the hotspot zone may borrow channels from the relatively cold cells (if any). However, a borrowed channel may be in use within the co-channel cell(s) of the lender cell. This causes channel interference and creates additional disturbances in information exchange. Therefore, during channel borrowing the necessary condition to be checked is that the lending cell has at least one free channel and that is not being used by any of its co-channel cell. Once borrowed, the channel in the lender as well as in all its co-channel cells are locked. Many variations of channel borrowing techniques are reported in the literature. However, it is still an active area of research. Detailed discussion on that is beyond the scope of this book.

Besides the manipulation of radio resources (such as channel), the other useful measures to increase the capacity of a system are 'cell splitting' and 'cell sectoring'. In 'cell splitting', the network cells are partitioned into smaller cells, whereas in 'cell sectoring', a cell is divided into several sectors resulting in increase in the system capacity.

2.10.1 Cell Splitting

The cell splitting is a mechanism of partitioning a network cell into a desired number of 'microcells', each having its own base station. When a

cell becomes congested, it is partitioned into smaller subcells (microcells), as illustrated in Figure 2.6, to increase the system capacity. The heavy-traffic regions (e.g. hotspots) are normally split into such smaller areas to ensure the acceptable grade of service.

In a microcell, antenna height as well as the transmitter power is lowered. While a cell diameter ($2R$) ranges from 2 to 20 kilometres, the microcell diameter ranges from hundred metres to a kilometre. The cell splitting technique decreases R (Figure 2.6) while leaving Q (frequency reuse ratio) relatively unchanged. This mechanism allows a system to grow by replacing large cells with smaller cells without upsetting the channel allocation scheme required to maintain the minimum frequency reuse ratio (Q or D/R).

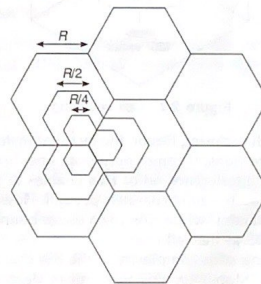


Figure 2.6 Cell splitting.

As an effect of cell splitting, the capacity of a network increases due to the additional number of channels per unit area. But the increased number of cells in a network adds more cell-boundaries. This may lead to more frequent handoffs. Therefore, the application of cell splitting is limited only to the cells with heavy traffic overloads. However, splitting only in a part of the system may result in serious channel assignment problems.

The designers favour cell splitting because of its capacity improving capability. The powers used by an MS and the BS after splitting is also comparatively low. The problem of unwanted handoff, however, is addressed with introduction of the concept of umbrella cells.

2.10.2 Cell Sectoring

Sectorization is the division of an omnidirectional (360°) view into non-overlapping slices called 'sectors' (Figure 2.7). Each sector has its own set of channels and considers directional antenna. The directional antennas at the BS are used to focus on the sectors. Usually this method divides a cell into three or six sectors. When a cell is sectorized, the R (cell radius) is remained

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unchanged. Further, D is reduced, and the amount of frequency reuse is increased. Hence the capacity is increased. Therefore, bandwidth efficiency of the system is enhanced as the frequency can be reused more often.

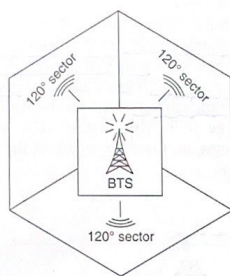


Figure 2.7 Cell sectoring.

The merit of cell sectoring lies on the fact that instead of interference received from all directions, it comes only from one direction causing the increase in signal to interference ratio. This enables to reduce the cluster size, and in effect allocates more channels per cell. However, within a cell an MS may need to handoff while crossing a sector boundary. The handoff is managed by the BS at the cell centre.

The cell sectoring adds complexity at the BSs due to introduction of additional antennas. Moreover, this mechanism decreases the trunking efficiency (queuing efficiency) while a large number of customers receive service from a set of servers. The proportional assignment of customers to different servers is a better solution.

2.11 USER VALIDATION IN CELLULAR COMMUNICATION

Just like any other network, user validation is an essential task in cellular network. In this section, the subscriber validation process, implemented in AMPS, is described.

2.11.1 Sources of Piracy

A common practice in unauthorized use of terminals is the cloning of stolen terminals. In AMPS, a terminal stores two ID numbers—the electronic serial number (ESN) and mobile identification number (MIN). The ESN is a 32-bit hardware-based serial number (unique 8-bit manufacturer's code + 24-bit ID for the mobile node for the given manufacturer's code). The MIN corresponds to a user telephone number (area code + phone ID)

assigned when a subscriber account is opened. Both the ESN and MIN are stored in the mobile terminal (MS).

The intruder can extract the ESN of a subscriber from the EEPROM (Electrically Erasable Programmable ROM) of the mobile device. The conventional measure for preventing unlawful access to such IDs is the encryption. The ESN is encrypted before writing them into the mobile terminal's EEPROM. Whenever the terminal tries to send ESN, the terminal decrypts the ESN first and then transmits over the air. That is, an ESN is easily accessible to the intruder who taps the network transmission. Therefore, it requires further elaborated security measure that prevents access to the network resources by an intruder. One such popular scheme is described in the next section.

2.11.2 Validation

The validation procedure in AMPS relies on the two IDs of a terminal (MS)—the ESN and MIN. The MS sends both the (ESN, MIN) to the network for:

- Registration
- Initiating a call
- Roaming

The subscriber validation process in the early AMPS system implemented verification of the transmitted tuple (ESN, MIN) with that already registered in the network. It also maintained the list of stolen terminal IDs to protect the system from piracy (Section 2.11.1). However, as both the IDs were transmitted straight over the air, an intruder had the scope to clone the ESN and MIN. Therefore, protection for unlawful access to the network was not guaranteed.

The advanced AMPS system uses a key-based authentication procedure to validate a subscriber. The serving network provides an A-key (64-bit primary secret key) value for each MS. During authentication a shared secret data (SSD, 128-bit shared secret key) is generated from the A-key and the ESN. The mobile terminal then runs an authentication algorithm and generates a terminal authentication result (MS-AUTHR). The MS-AUTHR along with the ESN and MIN is then transmitted from the MS to the network. Upon receipt of the information, the network registers the terminal. The serving network also computes SSD separately with the stored A-key and based on this SSD a network generated NT-AUTHR is produced. Finally, the NT-AUTHR and the MS-AUTHR are compared. If comparing a match indicates the MS is a valid user, the network allows the call initiated from the MS to proceed.

An unauthorised MS is not supposed to know A-key and hence the SSD is not known to it, the authentication data MS-AUTHR sent by the unauthorised MS will not match the NT-AUTHR. Therefore, the system should be able to recognise the unauthorised MS and denies its service. The

algorithmic steps of this authentication process (Figure 2.8) are noted below:

- An MS has a unique A-key supplied by the serving network.
- A SSD (shared secret key) is derived at the MS from A-key and ESN.
- MS runs an authentication algorithm and generates Authentication Result (MS-AUTHR) based on the SSD.
- The network produces a network generated AUTHR (NT-AUTHR) with the help of A-key stored in it.
- The NT-AUTHR and MS-AUTHR are compared at the network station. If there is a match, the network allows the call to proceed.

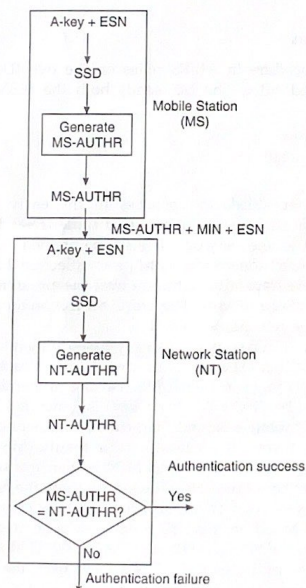


Figure 2.8 User validation process.

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REVIEW QUESTIONS

1. Define system capacity in the context of cellular mobile network.
2. What is cellular concept? How cellular concept enhances system capacity?
3. Describe cellular mobile network architecture. Which part of communication is wireless in such a network?
4. Explain the tasks of MS, BS, MSC, HLR and VLR.
5. What roles the channels FVC, RVC, FCC and RCC play in cellular mobile network?
6. Why hexagonal cell shape is normally used in cellular system design? In spite of the fact that hexagonal cell shape is most desirable, in practice, irregular cell shapes are considered. Why?
7. (a) Define frequency reuse. Why frequency reuse is essential?
(b) What is co-channel cell?
(c) How a compact pattern or cluster of cells is formed?
(d) How the nearest co-channel cells are identified?
8. What are co-channel interference and adjacent channel interference? How they can be reduced?
9. What is meant by near-far problem? What mechanism is employed to combat such a problem?
10. Show that the system capacity increases with the number of times the cluster replicates.
11. (a) Consider 40 MHz bandwidth is assigned to a cellular mobile network. The system uses two simplex channels of 20 kHz to provide full-duplex voice and control channels. Calculate the number of channels to be assigned per cell for a cluster of size 12.
(b) Consider a cellular system having 2023 duplex channels to cover 1925 sq. km and each cell area is 5 sq. km for 7-cell reuse system. Compute system capacity.
12. How the two parameters, system capacity and interference, play important roles in system design?
13. What is frequency reuse ratio? Derive the relationship between the frequency reuse ratio and signal-to-interference ratio.
14. When a BS receives signal from a communicating MS, state the causes of interference the signal may encounter.

15. Describe different channel assignment strategies.
16. Compare the following:
 - (a) NCHO vs. MAHO
 - (b) Hard handoff vs. soft handoff
 - (c) 1st generation vs. 2nd generation handoff
17. Does MAHO require any action taken by the MSC? If yes, what role the MSC plays in such a scheme?
18. Why handoff prioritization is essential? How is it implemented?
19. What is umbrella cell? Why the concept of umbrella cell is used?
20. What is roaming?
21. What are the issues involved in mobility management?
22. (a) Define GOS and Erlang.
 (b) What are Lost Calls Cleared (LCC) and Lost Calls Delayed (LCD)?
 (c) Write assumptions (if any) to implement LCC.
23. Without manipulating radio resources, what are the other mechanism to increase system capacity?
24. Write the full forms of ESN and MIN. How are they used for validating a subscriber?

3

CELLULAR NETWORK STANDARDS (GSM AND IS-95)

The cellular Mobile Systems of the early era were very much analog systems. These were mostly installed in vehicles and not at all prepared to be used as hand-held devices. The analog systems of that era had poor voice quality and suffered from interference of signals. Further, in analog system security of communication was found difficult to implement. These factors limited the massive use of mobile systems. The low-cost, hand-held mobile station with longer battery life, better quality of service, enhanced security was only possible after the introduction of digital communication systems. This chapter attempts to make the readers familiarized with the commercially available digital cellular communication standards such as GSM, IS-95.

3.1 DIGITAL CELLULAR COMMUNICATION

Digital radio was first introduced in the defence sector to ensure quality reception with a high level of security in an interference-prone zone. The demand of hand-held terminal with reduced size and power requirement, longer battery life, etc. has led to explore digital cellular system commercially. An improvement in VLSI technology coupled with the advancement in the Digital Signal Processing (DSP) has made it a reality. The digital cellular system outperforms an analog system in terms of:

- capacity
- quality of service
- security
- ability to support improved services such as wireless Internet
- battery life

Digital scrambled up the signals into bursts; so it is more secure than analog and, therefore, helps prevention of stealing phone account information. However, roaming is comparatively difficult with a digital phone than an analog. As there is no uniquely accepted industry standard in digital technology, roaming, i.e. using an operator's network other than the home network can be difficult. On the other hand, an analog system has better coverage and the initial cost for analog is usually less than digital. Therefore, to get the best voice quality as well as security, the terminal offering dual mode (digital/analog) feature can be a better choice. This allows automatic switching between the analog and digital modes depending on the controlling antennae of the region the mobile station (MS) currently belongs to.

3.2 MULTIPLE ACCESS TECHNIQUES

The frequency spectrum allocated to an application (e.g. cellular communication) is shared by the users in a digital system. The sharing mechanism is realized with any one of the multiple access techniques mentioned in Section 1.2.3.

In an effort to make the most efficient use of radio frequency spectrum, a finite natural resource, various technologies have been developed. The target is to simultaneously support as many users as possible by a finite range of spectrum. In a cellular network, transmission from the base station (BS) in forward (downlink) channel can be received by all the MSs under its control. This, in a word, is broadcasting. On the other hand, in the reverse (uplink) channel multiple users want to transmit information simultaneously to the serving base station (BS). Without proper coordination among the transmitting users, collisions occur. Therefore, when two or more users transmit simultaneously to the same serving base station, it needs to be conflict free. This necessitates the scheme for multiple accesses.

The key target in multiple accesses is to separate the transmitted signals from different users at the base station. Different techniques of user separation at receiver (BS) have given birth to different multiple access techniques, namely Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA) and Code Division Multiple Access (CDMA).

In FDMA, a user separation is implemented by providing separate frequency band to individual user whereas in TDMA, it is done by providing separate time slot to each user. On the other hand, CDMA uses separate code to differentiate transmitted signal from an individual user. The 1st generation wireless systems use FDMA while the 2nd generation systems use TDMA and CDMA.

3.2.1 FDMA

An FDMA system separates out the total available bandwidth into several non-overlapping smaller bands/channels. Each channel has the ability to support one user. Each user is assigned a unique frequency band or channel.

Figure 3.1 illustrates that each user is allowed a unique frequency band. These channels are assigned on demand. When an MS tries to communicate with a BS, it registers itself using control channel to the closest BS. During registration, the BS assigns the MS an available pair of channels, one to transmit (reverse channel) and the other to receive (forward channel).

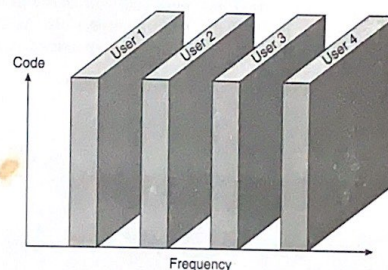


Figure 3.1 Frequency division multiple access.

To ensure interference-free transmission between two users or between the forward and reverse channel transmission of a user, it introduces the guard bands. The guard band separates out two such channels preventing interference. The guard bands are unused portions of the spectrum. For example, if 500 MHz bandwidth is provided for a system, entire bandwidth can be split into 12 channels each of 40 MHz. Each 40 MHz channel includes a 4 MHz guard channel. Effectively each channel is 36 MHz wide. In case of forward and reverse channels, either two antennas operating at different frequencies or one antenna with Frequency Division Duplexing (FDD) can provide the guard band.

As the multiple access is achieved by separating the user by the frequency allocated to them, frequency planning has to be done very carefully to avoid adjacent channel interference. This adjacent channel interference is an important factor in determining channel quality. However, frequency planning is complicated and difficult to achieve. Available frequency bands must be researched and analysed.

3.2.2 TDMA

In a system with a moderately large number of active users, the allocation of unique frequency band for each user cannot be a realistic option. As the demand increases to accommodate more users, spectrum efficiency¹ of

¹ Spectrum efficiency is measured as the amount of information that can be transmitted over a given bandwidth in a communication system. In the context of cellular mobile communication, spectrum efficiency means the maximum number of subscribers per cell that can be accommodated with an acceptable quality of service.

FDMA system becomes insufficient. A TDMA system splits users into an available pair of channels, but they also assign each user an available time slot within that channel. In each slot, only one user is allowed to either transmit or receive. Frequency division is still employed but these frequencies are now further subdivided into a defined number of time slots per frequency. Figure 3.2 shows that the entire spectrum is divided into four channels each of which is divided into three time slots. As the TDMA systems do not transmit all the time, the MSs gain an extended battery life as well as the talktime.

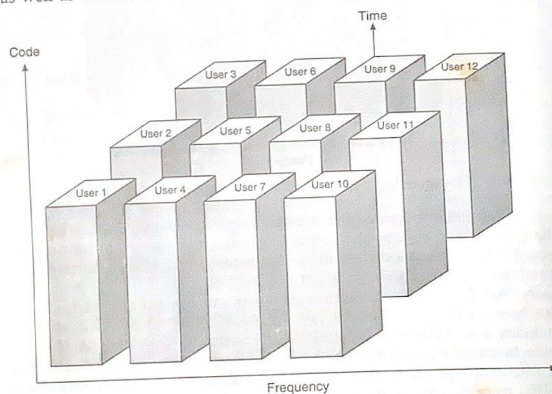


Figure 3.2 Time division multiple access.

In TDMA, like FDMA, the BS assigns the MS an available pair of channels. In addition to this, it assigns a time slot within the channel that must be available to the MS. The user can only be allowed to send or receive data for this time slot. The data bursts are fast reassembled at the receiving end, and therefore, appeared as continuous. Figures 3.1 and 3.2, point to the fact that while an FDMA system supports 4 users, the TDMA can support 12 users utilising the same bandwidth. As the TDMA systems first split an allotted portion of the frequency spectrum into smaller slots/channel, they require the same level of frequency planning as FDMA system.

3.2.3 CDMA

In code division multiple access (CDMA), signals from users (MSs) are differentiated at the receiver (BS) in code space whereas in FDMA it is

differentiated in frequency space and in TDMA, in time space. The main challenge of CDMA is to find suitable code for a user's signal so that it is unique and can be differentiated from the signals of other users. The hurdle is mitigated in CDMA by finding mutually orthogonal codes having high autocorrelation² value for each user. Two codes are orthogonal to each other when their cross correlation³ value is 0. However, the CDMA does not allocate unique channel for a user. Each user can utilize the entire block of allocated spectrum space to carry the messages, that is, all the users can use the same carrier frequency and transmit simultaneously.

There are two popular coding techniques used in CDMA—Walsh code⁴ and PN (pseudo noise) code⁵. In the forward (BS to MSs) direction, transmission for all the users originates from the same transmitter (BS) in a perfectly coordinated manner. The orthogonal Walsh code is used for forward channel transmission. In forward direction, transmission is originated from a BS, a fixed station. It enables the generation of orthogonal codes for the user signals. On the other hand, the reverse transmission (MS to BS) cannot be accurately coordinated due to the mobility of MSs. Generation of orthogonal codes for each user in such case is not at all possible for arbitrarily random locations of the MSs. Hence the PN code is used for the reverse channel transmission.

Spread spectrum

Spread spectrum is a technique that realizes the unique coding for signals from each user. This also improves the spectrum efficiency over that of FDMA and TDMA. The principle of spread spectrum communication is to spread the bandwidth of baseband information-carrying signals from different users to a much larger bandwidth. Ideally, the spreading signals used for different users are orthogonal to each other.

In spread spectrum, the narrow band information-carrying signal is multiplied by a very large bandwidth signal called the spreading signal. The bits of spreading signal are referred to as the chips. If T_c is the period of one chip, $1/T_c$ is the chip rate. The user data rate is $1/T_b$, where T_b is the period of one data bit (baseband information-carrying signal). The spreading factor is defined as the ratio of chip rate and user data rate (T_c/T_b).

One of the popular implementation of spread spectrum is the direct sequence spread spectrum (DSSS). In DSSS-based CDMA, the spreading signals with different chip rates are used to the different data signals. The chip rates used for the spreading signal are significantly greater than the

² Correlation is to determine how much similarity a code has with another code. It is computed by taking inner product of the codes. Autocorrelation is the correlation of a user code with itself.

³ Cross correlation is correlation between two separately generated codes.

⁴ Walsh codes are mathematically orthogonal codes.

⁵ A PN code is a binary sequence that appears randomly but can be reproduced in a deterministic manner.

user data rates. Normally, the chip rate is in the order of "megachips per second" (Mcps), that is, millions of chips per second. For DSSS-based CDMA modulation (Refer to Figure 3.3(a)), the first task is to spread the information-carrying (user data) signal (digital modulation) and the second task is to modulate spread signal with carrier frequency. For example, spreading a user signal of 1 MHz bandwidth with 11 chip code generates a signal of 11-MHz bandwidth. Then the signal is converted to GHz bandwidth to make it ready for transmission. Demodulation (Refer to Figure 3.3(b)), the reverse process of modulation, is employed to retrieve user data at the receiver (BS) side.

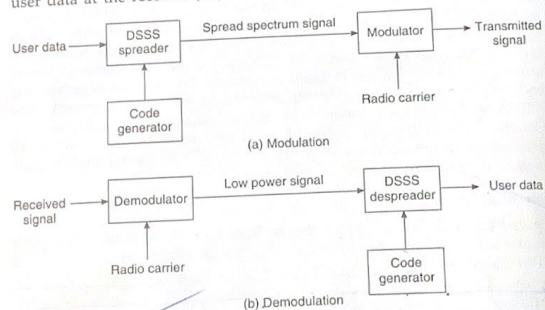


Figure 3.3 DSSS CDMA system.

If sufficient bandwidth is available, spreading is performed in two consecutive phases. In one phase data is spread by an orthogonal code to provide mutual orthogonality among all users in the same cell. In the next phase the resulting signal is further spread by a PN sequence to provide mutual orthogonality among the users in different cells. Such two phase spreading is called multiple spreading. IS-95 adopts this multiple spreading technique.

In CDMA for both the forward and reverse channel transmission, unique codes (walsh code and PN code respectively) are used for a user. A receiver would try to reconstruct only the specific desired user's data. And all other users' data would be sensed as noise. While detecting the user data signal, a receiver (BS) needs to know the code used by the transmitter (MS). An analogy can be drawn with a business party being held in a big hall. As it is shown in Figure 3.4, a number of persons are sharing views among themselves. Each pair of them is using a unique language for conversation. As a pair is using separate language (code), others conversations cannot be affected severely provided no one speaks much loudly compared to others. If one speaks so, it will add to noise.

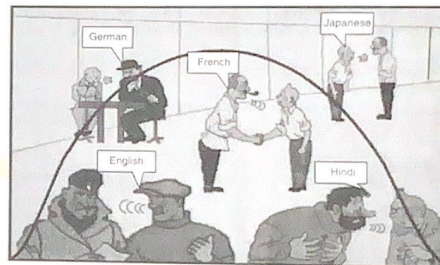


Figure 3.4 An analogy with CDMA.

It is mentioned earlier that the CDMA demands the codes should be orthogonal to each other and a code for a certain user should have a good autocorrelation. For example, let us consider the two simplest codes (1, 0) and (1, 1). Assuming 0 as -1 and 1 as +1, the codes become (+1, -1) and (+1, +1). The inner product of these two is $(+1) \cdot (+1) + (-1) \cdot (+1) = +1 - 1 = 0$ and hence the codes are orthogonal. A code with high autocorrelation means the absolute value of inner product of the code with itself is high. For example, for the code (+1, -1), the autocorrelation value is $(+1, -1) \cdot (+1, -1) = (+1) \cdot (+1) + (-1) \cdot (-1) = 1 + 1 = 2$. An example of a code, used for ISDN and IEEE 802.11 is 11 chip code (+1, -1, +1, +1, -1, +1, +1, -1, -1, -1, -1). The autocorrelation value of this code is 11.

How it works

Consider two MSs, MS_a and MS_b that want to transmit data_a (1, 0) and data_b (1, 1) respectively. CDMA assigns two unique and orthogonal codes—code_a = (0, 1, 0, 1), code_b = (0, 1, 1, 0). Both the MS_a and MS_b spread respective signal using their unique key as chip sequence (spreading signal). The spreading algorithm implies that if the data bit is 1, then send the corresponding code itself and for data bit 0, send the complement of the corresponding code. The resulting signals (encoded data_a and data_b) are given below. For convenience binary 0 is assumed as -1, and binary 1 as +1.

Encode MS_a
 $code_a = (-1, +1, -1, +1)$
 $data_a = (+1, -1)$
 $encoded\ data_a = code_a \cdot data_a$
 $= (-1, +1, -1, +1) \cdot (+1, -1)$
 $= (-1, +1, -1, +1)$
 $signal_a = (-1, +1, -1, +1, +1, -1, -1, +1, -1, +1)$

Encode MS_b
 $code_b = (-1, +1, +1, -1)$
 $data_b = (+1, +1)$
 $encoded\ data_b = code_b \cdot data_b$
 $= (-1, +1, +1, -1) \cdot (+1, +1)$
 $= (-1, +1, +1, -1)$
 $signal_b = (-1, +1, +1, -1, -1, +1, -1, +1, -1, +1)$

The received signal at the BS is the sum of these two signals; signal_a and signal_b , if there is no other MSs under the serving BS. That is, $\text{signal}_{\text{received-at-BS}} = \text{signal}_a + \text{signal}_b = (-2, +2, 0, 0, 0, 0, +2, -2)$. As the code_a and code_b are orthogonal to each other, the data_a and data_b can easily be extracted from the $\text{signal}_{\text{received-at-BS}}$. The decoding performed at BS to extract the data sent by the MS_a and MS_b are described below.

The extraction of data sent by MS_a means:

- (i) $\text{decode}_a = \text{signal}_{\text{received-at-BS}} \cdot \text{code}_a$
- (ii) replace values greater than 0 of decode_a as 1 and values less than 0 as 0 to get the data_a . Therefore

$$\begin{aligned} \text{Decode MS}_a \\ \text{code}_a &= (-1, +1, -1, +1) \\ \text{signal}_{\text{received-at-BS}} &= (-2, +2, 0, 0, 0, 0, +2, -2) \\ \text{(i) decode}_a &= \text{signal}_{\text{received-at-BS}} \cdot \text{code}_a \\ &= ((-2, +2, 0, 0), \\ &\quad (0, 0, +2, -2)) \cdot (-1, +1, -1, +1) \\ &= (2 + 2 + 0 + 0), (0 + 0 - 2 - 2) \\ &= (+4, -4) \\ \text{decode}_a &= (+4, -4) \\ \text{(ii) data}_a &= (1, 0) = (+1, -1) \end{aligned}$$

$$\begin{aligned} \text{Decode MS}_b \\ \text{code}_b &= (-1, +1, +1, -1) \\ \text{signal}_{\text{received-at-BS}} &= (-2, +2, 0, 0, 0, 0, +2, -2) \\ \text{(i) decode}_b &= \text{signal}_{\text{received-at-BS}} \cdot \text{code}_b \\ &= ((-2, +2, 0, 0), \\ &\quad (0, 0, +2, -2)) \cdot (-1, +1, +1, -1) \\ &= (2 + 2 + 0 + 0), (0 + 0 + 2 + 2) \\ &= (+4, +4) \\ \text{decode}_b &= (+4, +4) \\ \text{(ii) data}_b &= (1, 1) = (+1, +1) \end{aligned}$$

* (A CDMA system provides more privacy than FDMA or TDMA systems.) Due to the wide bandwidth of a spread-spectrum signal in CDMA, it is very difficult to cause jamming or interference. It appears as nothing more than a slight rise in the noise floor⁶ or interference level. In other technologies, the power of signal is concentrated in a narrower band and, therefore, easier to detect or decode.

Increase in number of users in a CDMA system linearly raises the noise floor. Thus there is no absolute limit on the number of users in CDMA. Rather, the system performance gradually degrades for all users as the number of users is increased. Hence the CDMA has a soft capacity limit⁷. Moreover, there is a problem of self-jamming⁸. Self-jamming arises while spreading sequences of different users are not exactly orthogonal resulting in one user disrupting the transmission to the other, and vice versa. Further the users of CDMA share the same channel. It may lead to the near-far problem (Section 2.5). To combat this problem, power control

⁶ Noise floor is the measure of sum of signal generated from all the noise sources.

⁷ Soft capacity limit is a limit of accommodating number of users beyond which system performance degrades to an unacceptable extent.

⁸ Transmission of radio signals that disrupts communications by decreasing the signal-to-noise ratio.

technique is employed in most CDMA implementations so that each MS under a BS coverage can provide the same signal level to the BS receiver.

The well-designed filters employed in FDMA ensure zero or a minimal spectral overlap whereas employment of slot synchronization in TDMA reduces timing jitter. This makes both FDMA and TDMA conflict-free multiple access schemes. On the other hand, CDMA is a spread spectrum technique. Although orthogonality between transmitted signals from different users is the key element of CDMA, in reality the generated wideband spreading functions are not truly orthogonal and cause some interference. Hence CDMA is interference limited multiple access strategy.

The following discussion summarizes the merits and demerits of the three multiple access techniques—FDMA, TDMA and CDMA.

FDMA

Merits

- simple to implement
- fairly efficient with a small population and when traffic is almost constant
- fewer bits are needed as overhead (synchronization, etc.) as compared to TDMA

Demerits

- frequency planning is difficult
- if a channel is not used by the designated user, it cannot be used by others to increase or share capacity—a wastage of resources

TDMA

Merits

- data transmission for users is not continuous leading to low battery consumption
- can accommodate much more users in the same spectrum space as compared to FDMA; capacity increases in high traffic areas
- can allocate different number of time slots per frame to different users. Thus bandwidth can be supplied on demand to different users based on priority.

Demerits

- as high synchronization overhead is required, receivers are to be synchronized for each data burst and, therefore, requires additional overheads in comparison to FDMA
- frequency guard bands lead to spectrum inefficiency
- security is much lower than that in CDMA
- frequency planning is critical
- the number of users accommodated is less than CDMA in the same spectrum space

CDMA
Merits

- best spectrum efficiency: capacity increases 8 to 10 times than that of an analog system and 4 to 5 times than the other digital systems, making it most useful in high traffic areas with a large number of users and limited spectrum.
- simplified frequency planning as all users utilize the same radio frequency spectrum
- random walsh codes enhance user privacy/security

Demerits

- base station equipment is expensive
- difficult to implement in comparison to FDMA and TDMA
- continuous power adjustment of MS is essential to avoid near-far problem
- self-jamming arises from the fact that the spreading sequences of different users are not exactly orthogonal

Legacy commercial telecommunication networks such as analog networks based on Advanced Mobile Phone System (AMPS) are built around the FDMA. Currently, there are several digital cellular standards available worldwide. The example of TDMA-based standards are Global System for Mobile (GSM), Digital Cellular Systems (DCS), etc. On the other hand, IS-95 is the CDMA-based standard. The details of GSM (TDMA) and IS-95 (CDMA) standards are provided in the following sections.

3.3 GSM

Global system for Mobile (GSM) is a 2nd generation cellular system. It was developed in the 1990s to solve the fragmentation problems of first cellular systems and introduced in Europe. By 1993, GSM and its technically equivalent offshoot, DCS (Digital Cellular System) 1800, were adopted outside Europe, too.

GSM is the world's first cellular system to specify digital modulation and network level architectures and services. It has two objectives—pan-European roaming and interaction with the ISDN. Roaming yields compatibility around Europe and interaction with the ISDN offers capability to extend the single-subscriber-line system to a multi-service system. System capacity was not a primary issue at the initial development phase, however, to keep pace with the rapid growth in cellular services, many revisions have been made to GSM.

3.3.1 System Architecture

The GSM consists of three major interconnected components—Base Station Subsystem (BSS), Network and Switching Subsystem (NSS) and Operation

Support Subsystem (OSS). A mobile station (MS) interacts with BSS through air interface. The important interfaces are U_m , A_{bis} and A . Figure 3.5 illustrates the block diagram of GSM system.

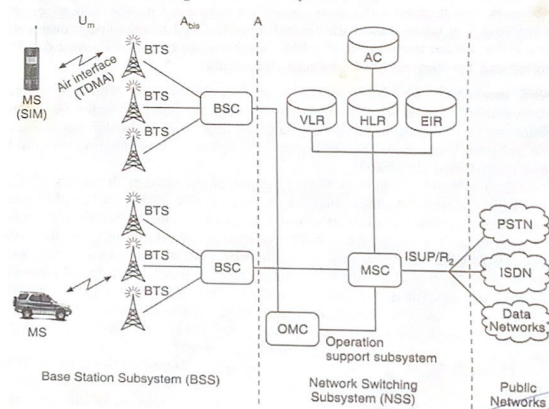


Figure 3.5 GSM architecture.

Mobile station: A mobile station comprises of device-dependent hardware/software and the subscriber-dependent SIM (subscriber identity module). The device is identified by IMEI (international mobile equipment identity) whereas a subscriber is identified by SIM. The device dependent hardware are transceiver, antenna, etc. and the softwares are for theft protection, etc. The SIM is a small detachable card and securely stores user specific information such as IMSI (international mobile subscriber identity), PIN (personal identity number), PUK (PIN unlocking key) and an authentication key to run user validation algorithm. The IMSI is similar to MIN whereas the IMEI is equivalent to ESN in AMPS. At any instant of time, an MS is either in idle state or in dedicated state. In the idle state, the MS only listens to the network but neither transmits nor receives any data. In this state it does not use any dedicated resources. On the other hand, in dedicated mode the MS transmits/receives data using dedicated resources. From an idle state, if an MS desires to access the network, it enters into the dedicated state and starts using resources.

BSS modules: A BSS consists of Base Station Controllers (BSCs), each of which controls a number of Base Transceiver Stations (BTSs). The BSCs